Bundesanstalt für Materialforschung und -prüfung (BAM)

Best Practice: Digital Reference Material Document (DRMD)

Guidelines for Standardized Implementation and Data Integrity

Status: First Draft

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Document Info	Details
Document ID	BAM-DRMD-BP-2026-001
Version	1.0
Release Date	May 2026
Classification	Public Release
Review Cycle	Annual

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Document Control & Revision History

Version	Date	Description of Changes	Author/Reviewer
1.0	May 8, 2026	Initial Release	

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Citation Suggestion

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Abbreviations & Glossary

BAM	Bundesanstalt für Materialforschung und -prüfung (Federal Institute for Materials Research and Testing).
BIPM	International Bureau of Weights and Measures (*Bureau International des Poids et Mesures*), providing the unit identifiers used in D-SI.
CAS	Chemical Abstracts Service (specifically referring to CAS Registry Numbers for chemical substances).
CRM	Certified Reference Material; signifies that values are certified with full traceability and uncertainty budgets.
DCC	Digital Calibration Certificate; the namespace and schema providing common data elements shared across digital certificate standards.
DOI	Digital Object Identifier; a persistent identifier used for indexing and global identification.
DRMD	Digital Reference Material Document; a standardized XML-based format designed to digitally represent reference material certificates in compliance with ISO 17034 and ISO 33401.
D-SI	Digital System of Units; a machine-readable representation of the International System of Units (SI).
ELN	Electronic Lab Notebook.
ERP	Enterprise Resource Planning system.
ISO	International Organization for Standardization.
LIMS	Laboratory Information Management System.
OES	Optical Emission Spectrometry.
RM	Reference Material; used for products where data is for information only and not used as a primary anchor for calibration.
RMP	Reference Material Producer; the organization responsible for producing the material and its documentation.
SDS	Safety Data Sheet.
SOP	Standard Operating Procedure.
TSP	Trust Service Provider; an entity that issues electronic seals or signatures.
UUID	Universally Unique Identifier; a mandatory string serving as a document's digital fingerprint to prevent ID collisions.
XML	eXtensible Markup Language; the technical format for DRMD certificates.
XMLDSig	XML Digital Signature; the standard used to ensure document authenticity and integrity.
XRF	X-ray Fluorescence.

XSD XML Schema Definition; used for structural validation of the document.

1 Introduction & Overview

1.1 Purpose & Scope

The Digital Reference Material Document (DRMD) is a standardized XML-based format designed to digitally represent reference material certificates in compliance with ISO 17034 (Competence of reference material producers) and ISO 33401 (Reference materials - Guidance for characterization and assessment of homogeneity and stability).

This best practice document provides comprehensive guidelines for machine manufacturers, reference material producers, and software developers to ensure consistent, interoperable, and automated handling of DRMD certificates across different platforms and applications.

The DRMD schema enables:

- **Automated certificate parsing and validation** without manual data entry.
- **Seamless integration** into laboratory information management systems (LIMS) and quality management software.
- **Standardized data exchange** between reference material producers and end-users.
- **Long-term digital preservation** of reference material metadata and measurement data.
- **Regulatory compliance** through structured documentation of traceability and uncertainty information.

1.2 Target Audience

Machine Manufacturers & Software Developers: Organizations developing applications for reference material management, calibration software, analytical instrumentation control systems, and laboratory information management systems (LIMS) that need to import and process DRMD certificates.

Reference Material Producers: Organizations certified according to ISO 17034 who generate and distribute DRMD certificates; these best practices guide in data structuring and certificate generation to ensure maximum interoperability.

End-Users & Laboratories: Analytical laboratories, calibration facilities, and quality control departments that load DRMD certificates into their instrumentation and software systems and require understanding of certificate structure for proper interpretation.

IT and Quality Assurance Personnel: Technical staff responsible for implementing DRMD-compatible systems, conducting data validation, ensuring data integrity, and maintaining audit trails for regulatory compliance.

1.3 Document Objectives

This best practice document aims to:

- **Eliminate ambiguity** in DRMD schema interpretation by providing detailed implementation guidance with real-world examples.
- **Establish data quality standards** for mandatory, recommended, and optional fields to ensure consistency across all DRMD instances.
- **Define validation requirements** that software must implement to verify DRMD certificate integrity, completeness, and compliance.
- **Provide a technical implementation roadmap** for software developers including XML parsing, namespace handling, and data type conversions.
- **Ensure interoperability** between different software platforms through standardized handling of common scenarios.
- **Support regulatory traceability** by documenting how DRMD captures metrological traceability chains, measurement uncertainty, and responsible person information.
- **Enable automated workflows** such as automatic certificate verification, property extraction, and integration into analytical methods.
- **Facilitate digital preservation** by establishing guidelines for long-term storage, versioning, and backward compatibility of DRMD documents.

2 DRMD Schema Overview & Architecture

2.1 Schema Structure & Six Core Containers

The DRMD schema is organized into six interconnected containers, each serving a specific function in representing the complete reference material certificate:

2.1.1 Administrative Data

The Administrative Data captures essential document metadata and organizational information required for certificate identification and traceability. This container includes core identification data (document title, unique identifier, validity period), producer information (name, contact details, location), and responsible person details (roles, signatures, designations) to establish clear accountability and authority for certificate issuance.

2.1.2 Materials

The Materials defines the reference materials covered by the certificate, including material name, detailed description, material class and identifiers, minimum sample sizes required for use, and item quantity information to enable users to understand what material they are working with and how much is needed for proper use.

2.1.3 Material Properties

The Material Properties documents the certified and informational properties of the material, including property names, measurement procedures, results (values, units, measurement uncertainty), `isCertified` attribute (true for certified values, false for informational values), and metrological traceability to ensure users can identify which properties are officially certified and understand the precision and reliability of each property value.

2.1.4 Statements

The Statements provides essential non-numerical information including intended use guidelines, storage requirements, handling instructions, metrological traceability statements, health and safety information, legal notices, subcontractor list, and reference to detailed certification reports to guide users on proper material application, handling, and storage.

2.1.5 Comments & Documents

The Comments & Documents allows for supplementary information through free-text comments (supporting notes on methodology, material history, measurement anomalies) and embedded or linked external documents (e.g., original PDF certificates, detailed technical reports, supplementary data files) to provide additional context.

2.1.6 Digital Signature

The Digital Signature uses XMLDsig elements (`ds:SignatureType`) to provide cryptographic digital signatures that guarantee the authenticity, integrity, and non-repudiation of the DRMD certificate, ensuring that the certificate has not been tampered with and came from the stated producer.

2.2 XML Namespace & Technical Specifications

DRMD certificates are XML documents that utilize multiple namespaces to organize different types of information:

drmd namespace (`https://example.org/drmd`): Contains the primary DRMD-specific elements for document structure, material properties, and reference material information.

dcc namespace (`https://ptb.de/dcc`): Provides common data elements shared across multiple dcc standards including names, contacts, descriptions, and organizational information.

si namespace (`https://ptb.de/si`): Handles scientific and metrological data including numerical values and units (often expressed as LaTeX strings).

ds (XMLDSig) namespace (`http://www.w3.org/2000/09/xmldsig`): Defines the standard XML Digital Signature elements used to sign the DRMD document and enable authenticity and integrity verification.

Key Technical Characteristics:

- **XML Version:** 1.0 with UTF-8 encoding.
- **Schema Validation:** All DRMD instances MUST be valid according to the XSD schema (`schemaVersion` attribute indicates compatibility).
- **Element Naming:** Use `camelCase` for element names; namespace prefixes MUST match schema definitions.
- **Data Types:** Strict adherence to defined data types (string, decimal, integer, boolean, date, URI) to ensure proper parsing by software.

2.3 Schema Version & Backward Compatibility

The DRMD schema includes a `schemaVersion` attribute (currently "0.3.0") on the root element to indicate the schema specification version. Software implementations MUST include version checking logic:

Major version changes (e.g., 0.x to 1.x): May introduce breaking changes requiring schema mapping or manual intervention.

Minor version changes (e.g., 0.3 to 0.4): Typically add optional elements and maintain backward compatibility.

Patch version changes (e.g., 0.3.0 to 0.3.1): Address bugs or clarifications without structural changes.

Software should implement version compatibility strategies such as documenting supported schema versions and providing migration pathways for older certificates.

3 Administrative Data

This section provides a definitive guide to the Administrative Data block of the Digital Reference Material Document (DRMD). This block serves as the "Digital Container" for the material, ensuring that any software or analytical instrument can verify the document's authenticity, its producer, and its valid lifespan before processing technical data.

3.1 Core Data (`coreData`)

The `coreData` pillar contains the primary metadata that identifies the document and its legal status.

3.1.1 Title of the Document (`titleOfTheDocument`)

Description: A mandatory classification of the document type.

Purpose: To define the metrological level of the material.

Role: It tells the analytical instrument's software how to weigh the data.

CRM Perspective: Selecting `referenceMaterialCertificate` signifies that the values are certified with full traceability and uncertainty budgets.

RM Perspective: Selecting `productInformationSheet` signifies that the data is for information only and should not be used as a primary anchor for calibration.

XML Examples:

```xml
<!-- Way 1: For a Certified Reference Material (CRM) -->
<drmd:titleOfTheDocument>referenceMaterialCertificate</drmd:titleOfTheDocument>

<!-- Way 2: For an informative Reference Material (RM) -->
<drmd:titleOfTheDocument>productInformationSheet</drmd:titleOfTheDocument>
```

3.1.2 Unique Identifier (`uniqueIdentifier`)

Description: A mandatory string that serves as the document's fingerprint.

Purpose: To prevent "ID collisions" in global databases.

Role: Used by Laboratory Information Management Systems (LIMS) to track version control. If a certificate is updated, a new UUID is issued.

XML Example:

```xml
<drmd:uniqueIdentifier>84d58aee-4f81-460f-993e-a594ce3a8d60</drmd:uniqueIdentifier>
```

3.1.3 Document Identifiers (`documentIdentifiers`)

The `documentIdentifiers` section acts as the digital "Linkage Hub" for the certificate, allowing it to be indexed and verified across different global and internal databases.

Core Components & Roles

`documentIdentifiers` **(List Container):** Serves as a "Master Folder" that bundles multiple external identities (e.g., DOI, Batch ID, ERP ID). It is optional, but if included, must contain at least one identifier.

`documentIdentifier` **(The Entry):** Represents one specific record in an external system. Contains mandatory technical details (scheme, value) and the optional actionable link.

`scheme` **(The Authority):** Defines the naming system or database being used (e.g., "DOI", "Internal_Warehouse_System"). It tells software how to interpret the identifier.

`value` **(The Key):** The actual unique code assigned by the authority. Primary data used for searching or matching in external databases.

`link` **(The Bridge):** Optional URI/URL leading directly to the source for immediate verification of origin.

Technical Attributes

`@id`**:** An internal XML label used to point other parts of the document to this ID.

`@refId`**:** A pointer used to link this ID to another ID elsewhere in the document.

`@refType`**:** Categorizes the nature of the ID (e.g., `basic_certificateIdentification`).

XML Example:

```xml
<drmd:documentIdentifiers>
  <!-- Global DOI Identification -->
  <drmd:documentIdentifier id="ID_01" refType="basic_certificateIdentification">
    <drmd:scheme>DOI</drmd:scheme>
    <drmd:value>10.1234/BAM.M308a.2026</drmd:value>
    <drmd:link>https://doi.org/10.1234/BAM.M308a.2026</drmd:link>
  </drmd:documentIdentifier>

  <!-- Internal Business Tracking -->
  <drmd:documentIdentifier id="ID_02">
    <drmd:scheme>Internal_ERP</drmd:scheme>
    <drmd:value>BATCH-2026-X99</drmd:value>
  </drmd:documentIdentifier>
</drmd:documentIdentifiers>
```

3.1.4 Validity (`validity`)

The Validity section is a mandatory choice that establishes the operational life of the certificate. Producers must select exactly one of the three following methods to ensure quality control and prevent the use of degraded materials.

`specificTime` **(Fixed Expiration):**

Description: A hard expiration date using the YYYY-MM-DD format.

Purpose: Sets a non-negotiable end date for the material's data.

Role: Provides a "Hard Stop" for safety. Automated instruments compare this to the current date and "lock out" the material if it has expired.

```xml
<drmd:validity>
  <drmd:specificTime>2040-12-31</drmd:specificTime>
</drmd:validity>
```

`timeAfterDispatch` **(Relative Validity):**

Description: A dynamic duration defined by an optional `dispatchDate` and a mandatory ISO 8601 period (e.g., P3Y for 3 years).

Purpose: Accommodates materials where shelf-life only begins once shipped to the customer.

Role: Enables Automated Calculation. Systems read the period and date to automatically generate a unique expiration date for that specific unit.

Policy option (strict signature-bound issuance): DRMD's legal issuance is defined as the act of signing (not printing/publishing/dispatching), then set dispatchDate to the date of the signing event that constitutes issuance. Only do this when signing time is explicitly your issuance time by regulation/process.

```xml
<drmd:validity>
  <drmd:timeAfterDispatch>
    <drmd:dispatchDate>2026-04-20</drmd:dispatchDate>
    <drmd:period>P3Y2M</drmd:period> <!-- Valid for 3 years, 2 months -->
  </drmd:timeAfterDispatch>
</drmd:validity>
```

`untilRevoked` **(Permanent Until Notice):**

Description: A boolean element fixed to the value `true`.

Purpose: Used for inherently stable materials that do not require a predetermined expiration date.

Role: Acts as a "Permanent" flag. The certificate remains active until a revocation or updated version is issued.

```xml
<drmd:validity>
  <drmd:untilRevoked>true</drmd:untilRevoked>
</drmd:validity>
```

3.2 Reference Material Producer

Path: /drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:referenceMateria
Type: drmd:referenceMaterialProducerType
Cardinality: Required (1..1) inside drmd:administrativeData

drmd:referenceMaterialProducer identifies the **organization legally and operationally responsible** for producing (or issuing) the reference material and/or its documentation. In many contexts, this is the CRM/RM producer (e.g., BAM, NIST, IRMM), but it can also represent the issuing body for the document if different from the physical producer.

Why it matters

Defines who is responsible for the RM identity and claims.

Provides contact points for clarifications, updates, complaints, or revocation.

Links the producer to external registries via unique identifiers.

Who uses it

- **RM Producer / Issuing Organization:** Provides authoritative identity and enables consistent re-use of metadata across many DRMDs.
- **Laboratories / Customers:** Use it to verify the issuer and find official contact details to confirm provenance.
- **Software Developers:** Use it to render the "Producer block" consistently and use identifiers as stable keys for deduplication.
- **Machine / Instrument Manufacturers:** Link results to producer identity for QA workflows and LIMS integration.
- **Auditors / Regulators:** Use it to validate issuer identity and support regulatory compliance.

3.2.1 Name

Path: /.../drmd:referenceMaterialProducer/drmd:name
Type: dcc:textType
Mandatory: Required (1..1)

A multilingual "human-friendly" name of the producer. As a `dcc:textType`, it is a container that holds one or more `dcc:content` elements (type `dcc:stringWithLangType`), where the `@lang` attribute follows ISO-639-1.

Best practices:

- Provide at least one `dcc:content`.
- For international publications, provide multiple languages (e.g., `en`, `de`).
- Use the official long form; avoid abbreviations unless included separately.

```xml
<drmd:name>
  <dcc:content lang="en">Federal Institute for Materials Research and Testing (BAM)</
      dcc:content>
  <dcc:content lang="de">Bundesanstalt fuer Materialforschung und -pruefung (BAM)</
      dcc:content>
</drmd:name>
```

3.2.2 Contact

Path: /.../drmd:referenceMaterialProducer/drmd:contact
Type: `dcc:contactType`
Mandatory: Required (1..1)

A structured contact block essential for requesting value clarifications, handling instructions, or additional documentation (reports, methods). It also enables automated systems to generate printable certificate headers.

Contact Name (Required)

Path: /.../dcc:contact/dcc:name | **Type:** `dcc:textType`

Use for the department or team, e.g., "CRM Sales" or "Customer Support".

```xml
<dcc:name>
  <dcc:content lang="en">BAM Sales / CRM</dcc:content>
</dcc:name>
```

Email / Phone / Fax / Link (Optional)

Paths: /dcc:eMail, /dcc:phone, /dcc:fax, /dcc:link

Best practice: Always provide at least an email or phone for accessibility.

```xml
<dcc:eMail>sales.crm@bam.de</dcc:eMail>
<dcc:phone>+49 30 8104 2061</dcc:phone>
<dcc:fax>+49 30 8104 72061</dcc:fax>
<dcc:link>https://www.bam.de/</dcc:link>
```

Location (Required)

Path: /.../dcc:contact/dcc:location | **Type:** `dcc:locationType`

A repeating choice of address components (`street`, `postCode`, `city`, etc.). `dcc:countryCode` (ISO 3166-1) is highly recommended.

```xml
<dcc:location>
  <dcc:street>Richard-Willstaetter-Str.</dcc:street>
  <dcc:streetNo>11</dcc:streetNo>
  <dcc:postCode>12489</dcc:postCode>
  <dcc:city>Berlin</dcc:city>
  <dcc:countryCode>DE</dcc:countryCode>

  <dcc:positionCoordinates>
    <dcc:positionCoordinateSystem>WGS84</dcc:positionCoordinateSystem>
    <dcc:positionCoordinate1>
      <si:label>latitude</si:label>
      <si:value>52.4319</si:value>
      <si:unit>deg</si:unit>
    </dcc:positionCoordinate1>
    <dcc:positionCoordinate2>
      <si:label>longitude</si:label>
      <si:value>13.5330</si:value>
      <si:unit>deg</si:unit>
    </dcc:positionCoordinate2>
  </dcc:positionCoordinates>
</dcc:location>
```

3.2.3 descriptionData (Optional Attachment)

Path: /.../drmd:contact/dcc:descriptionData
Type: `dcc:byteDataType`

Used to attach small binary artifacts such as a vCard (`text/vcard`) or an organization logo (`PNG/SVG`).

```xml
<dcc:descriptionData>
  <dcc:fileName>producer-contact.vcf</dcc:fileName>
  <dcc:mimeType>text/vcard</dcc:mimeType>
  <dcc:dataBase64>BASE64_ENCODED_BYTES_HERE==</dcc:dataBase64>
</dcc:descriptionData>
```

3.2.4 Cryptographic capability flags (optional booleans)

Elements:

Path: /.../drmd:referenceMaterialProducer/drmd:cryptElectronicSeal
Type: `xs:boolean` (Optional)

Path: /.../drmd:referenceMaterialProducer/drmd:cryptElectronicSignature
Type: `xs:boolean` (Optional)

Path: /.../drmd:referenceMaterialProducer/drmd:cryptElectronicTimeStamp
Type: `xs:boolean` (Optional)

What they mean
These flags are **capability / policy hints**, not proofs. They help consumers know what to expect:

- **Electronic seal:** Organization-level sealing (often for automated issuing).
- **Electronic signature:** Person-based signing workflows.
- **Electronic timestamp:** Time-stamping services used for long-term validity.

Who uses them

- **Producers:** To declare available trust mechanisms.
- **Consumers:** To decide if additional manual verification is needed.
- **Software developers:** To decide whether to show validation UI and "digitally signed" expectations.

Best practice:

- Use `true/false` explicitly only when you are confident.
- Omit the element if unknown rather than guessing.

```
<drmd:cryptElectronicSeal>true</drmd:cryptElectronicSeal>
<drmd:cryptElectronicSignature>false</drmd:cryptElectronicSignature>
<drmd:cryptElectronicTimeStamp>true</drmd:cryptElectronicTimeStamp>
```

3.2.5 Organization Identifiers (optional list)

Element:

Path: /.../drmd:referenceMaterialProducer/drmd:organizationIdentifiers
Type: `drmd:organizationIdentifierListType` (Optional)

Contents
This is a list of `drmd:organizationIdentifier` elements, each containing:

- `drmd:scheme` **(Required):** Defines the naming system (e.g., ROR, VAT).

- `drmd:value` **(Required):** The actual unique code.
- `drmd:link` **(Optional):** A canonical resolver URL.
- **Attributes:** `@id`, `@refId`, and `@refType` (e.g., `basic_certificateIdentification`).

What it is
A mechanism to assign **stable external identifiers** to the organization, enabling unambiguous machine linking across global systems.

Recommended schemes (examples)

- **ROR:** Research Organization Registry ID.
- **VAT:** EU VAT number.
- **LEI:** Legal Entity Identifier.
- **National registers:** Company register or authority IDs.

Best practice:

- Prefer globally resolvable IDs (e.g., ROR + link).
- Use `link` whenever there is a canonical resolver URL.
- Keep the `scheme` short and stable; do not use long prose.

```xml
<drmd:organizationIdentifiers>
  <drmd:organizationIdentifier>
    <drmd:scheme>ROR</drmd:scheme>
    <drmd:value>02h2x7d13</drmd:value>
    <drmd:link>https://ror.org/02h2x7d13</drmd:link>
  </drmd:organizationIdentifier>

  <drmd:organizationIdentifier>
    <drmd:scheme>VAT</drmd:scheme>
    <drmd:value>DE123456789</drmd:value>
  </drmd:organizationIdentifier>
</drmd:organizationIdentifiers>
```

3.3 Responsible Persons

The Responsible persons block records the **human (or organizational) accountability** for the DRMD: who prepared it, who approved it, and if applicable who performed or is authorized to perform an electronic signature/seal/time-stamp. This is important for:

- **Reference Material Producer (RMP):** Governance, traceability of approvals, auditability, and responsibility assignment.

- **Regulators / Accreditation Bodies / Auditors:** Clear accountability and roles.
- **Software Developers / Platform Operators:** Consistent identity anchors (`@id`) and re-use via references (`@refId`) across sections; signature workflow flags.
- **Downstream Users (Labs, Customers):** Who to contact for clarifications and who formally endorses the document.

Path (in DRMD): `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:respP`
Type: `dcc:respPersonListType` (imported from DCC schema)

3.3.1 Responsible persons – list

Path: `/.../drmd:administrativeData/drmd:respPersons`

What it is
A required container holding one or more `dcc:respPerson` entries.

Why it matters

Ensures the DRMD always has at least one accountable person/entity.

Allows multiple roles (e.g., technical reviewer, quality manager, authorized signer).

How to use

Include **at least one** responsible person.

Include **multiple** persons when responsibilities are split (technical + quality + authorization).

Keep the list stable across document revisions (same people may re-appear); use `@id` to make them referenceable.

Cardinality
`drmd:respPersons` is **mandatory** in DRMD (`minOccurs=1` at `administrativeDataType` level). Inside it: multiple `dcc:respPerson` are allowed (DCC defines it as an unbounded list).

3.3.2 Responsible person – entry

Path: `/.../drmd:respPersons/dcc:respPerson`

What it is
One responsible person record.

Core purpose
Capture **who** is responsible (person/contact details) and optionally **what role** they play and **whether** they are the main signer.

Attributes (identity & linking)
These attributes enable stable identifiers and cross-references (important for larger DRMDs, signatures, or reuse across sections).

`@id` (xs:ID, optional)

- **Purpose:** Provides a unique identifier for this `respPerson` within the XML document.
- **Best practice:** Always set it (e.g., `respPerson_1`, `qm_1`, `signer_main`) so other elements can reference it consistently.

`@refId` (xs:IDREFS, optional)

- **Purpose:** References one or more IDs defined elsewhere (or earlier) in the document.
- **Best practice:** Use when you want to point to an existing identity block to avoid duplication (especially if the same person appears in multiple contexts).

`@refType` (dcc:refTypesType, optional)

- **Purpose:** Classifies the reference relationship(s).
- **Best practice:** Use controlled values defined by your profile/governance (this element uses `dcc:refTypesType` because `respPerson` is in the DCC namespace/type).

Child elements (accountability & signature workflow hints)

`dcc:person` (type: `dcc:contactNotStrictType`, required)

- **Purpose:** The responsible person's identity and contact data. "NotStrict" means location is optional (unlike `dcc:contactType` where location is required).
- **Best practice:** Provide at least name; include email/phone if this person is an operational contact.

`dcc:description` (type: `dcc:richContentType`, optional)

- **Purpose:** Human-readable description of responsibilities, expertise, department, scope, etc.
- **Best practice:** Use for role context ("Quality management", "Certification lead", "Technical reviewer").

`dcc:role` (type: `dcc:notEmptyStringType`, optional)

- **Purpose:** Short role label.
- **Best practice:** Keep it concise and consistent (e.g., "Head of Department 1", "Quality Manager", "Authorized Signer").

`dcc:mainSigner` (xs:boolean, optional)

- **Purpose:** Flags the primary signer among multiple persons.
- **Best practice:** If you have a signing workflow, set exactly one `mainSigner=true`.

`dcc:cryptElectronicSeal` / `Signature` / `TimeStamp` **(xs:boolean, optional)**

- **Purpose:** Indicate which cryptographic functions are relevant/expected for this responsible person in your workflow (seal/signature/timestamp).
- **Best practice:** Use these as workflow hints (not legal proof by themselves). If your system supports e-signature/seal/timestamp, set them consistently with your signing policy.

3.3.3 Person Details

Path: `/.../dcc:respPerson/dcc:person`
Type: `dcc:contactNotStrictType` (Required)

Child elements

`dcc:name` **(type:** `dcc:textType`**, required)**

- **Purpose:** Person's name as multilingual structured text.
- **Best practice:** Provide at least one `dcc:content`. Add `@lang` for multilingual versions.

`dcc:eMail` **(type:** `dcc:notEmptyStringType`**, optional)**

- **Purpose:** Email contact.

`dcc:phone` **(type:** `dcc:notEmptyStringType`**, optional)**

- **Purpose:** Phone contact.

`dcc:fax` **(type:** `dcc:notEmptyStringType`**, optional)**

- **Purpose:** Fax contact (legacy, but schema-supported).

`dcc:link` **(type:** `xs:anyURI`**, optional)**

- **Purpose:** Link to profile page, ORCID page, institutional directory entry, etc.

`dcc:location` **(type:** `dcc:locationType`**, optional in** `contactNotStrictType`**)**

- **Purpose:** Optional address/affiliation location.

`dcc:descriptionData` **(type:** `dcc:byteDataType`**, optional)**

- **Purpose:** Binary attachment (e.g., vCard, signed mandate PDF, authorization letter, portrait, etc.).
- **Best practice:** Use sparingly; if large, prefer external references in `link` unless your policy requires embedding.

How this structure helps different stakeholders

- **Producer:** Internal governance and audit.

- **Machine/Software Vendors:** Can parse and display a clear "Responsible persons" section; can route questions or support requests.
- **Customers/Labs:** Know who approved/certified the DRMD and how to reach them.

Example: Responsible Persons

```xml
<drmd:respPersons xmlns:drmd="https://example.org/drmd"
                  xmlns:dcc="https://ptb.de/dcc">

  <dcc:respPerson id="respPerson_1" refType="basic_certificateIdentification">
    <dcc:person>
      <dcc:name>
        <dcc:content lang="en">Dr. F. Emmerling</dcc:content>
      </dcc:name>

      <dcc:eMail>f.emmerling@example.org</dcc:eMail>
      <dcc:phone>+49 30 8104 0000</dcc:phone>

      <dcc:link>https://www.example.org/staff/emmerling</dcc:link>
    </dcc:person>

    <dcc:description>
      <dcc:content lang="en">Analytical Chemistry; Reference Materials</dcc:content>
    </dcc:description>

    <dcc:role>Head of Department 1</dcc:role>

    <dcc:mainSigner>true</dcc:mainSigner>

    <dcc:cryptElectronicSeal>true</dcc:cryptElectronicSeal>
    <dcc:cryptElectronicSignature>true</dcc:cryptElectronicSignature>
    <dcc:cryptElectronicTimeStamp>true</dcc:cryptElectronicTimeStamp>
  </dcc:respPerson>

  <dcc:respPerson id="respPerson_2"
      refId="respPerson_1"
      refType="basic_certificateIdentification">
    <dcc:person>
      <dcc:name>
        <dcc:content lang="en">Dr. S. Recknagel</dcc:content>
      </dcc:name>

      <dcc:eMail>sebastian.recknagel@example.org</dcc:eMail>
      <dcc:phone>+49 30 8104 1160</dcc:phone>
    </dcc:person>

    <dcc:description>
      <dcc:content lang="en">Specialist in Inorganic Reference Materials</dcc:content>
    </dcc:description>

    <dcc:role>Head of Division 1.6</dcc:role>
```

```xml
    <dcc:mainSigner>false</dcc:mainSigner>

    <dcc:cryptElectronicSignature>true</dcc:cryptElectronicSignature>
    <dcc:cryptElectronicTimeStamp>true</dcc:cryptElectronicTimeStamp>
  </dcc:respPerson>

</drmd:respPersons>
```

4 Materials

Path: `/drmd:digitalReferenceMaterialDocument/drmd:materials`
Contains one or more `drmd:material` elements.

What is covered in this section:
The Materials section is the authoritative catalogue of the physical reference material(s) that the DRMD describes. It answers, in a machine-readable way:

- **What material(s) exist** in this document (names, descriptions).
- **How they are classified** (optional repeatable material classes).
- **How they are identified externally** (IDs like BAM/CRM code, batch/lot, ROR, internal IDs, etc.).
- **What sample amount(s) are relevant** for using the material (e.g., minimum sample size; optional additional item quantities).

4.1 Purpose / use

Reference Material Producer (RMP):

- Publishes the canonical product identity (name + identifiers) and how to handle it (sample size).
- Ensures traceability between the digital document and physical packaging/labels, catalog entries, and internal inventory systems.

Laboratories / end users:

- Know exactly which material the certified values apply to (especially crucial when there are multiple materials, lots, forms).
- Determine minimum sample size to plan method suitability, sample preparation, and replicate strategy.

Instrument / machine manufacturers:

- Can ingest standardized material metadata to preconfigure methods.
- Map "material name/identifier" to internal instrument libraries.
- Display correct sample requirements and warnings in UI.

Software developers / LIMS / ELN / data platforms:

- Use `materials/material/materialIdentifiers` to match DRMD content to database entities (product tables, lots, customer inventory).
- Use DCC quantity structures to reuse existing parsing/validation for numeric values and units.

- **Handling Multiple Materials:** When a DRMD contains multiple `drmd:material` entries, the identifiers declared in:
  `/drmd:materials/drmd:material/materialIdentifiers`
  should be treated as the **canonical keys** for linking downstream information.
- In the next chapter (Material Properties), software can use these material identifiers as stable references (e.g., via `@id/@refId` patterns) to associate each set of properties with the correct material.

Regulators / auditors / accreditation bodies:

- Verify the document refers to a clearly identified material, and the declared minimum sample size and identifiers are consistent with external references and packaging.

4.2 Materials – List

Element: `drmd:materials`
Path: `/drmd:digitalReferenceMaterialDocument/drmd:materials`
Type: `drmd:materialListType`
Cardinality: required (exactly 1 at root)
Contents: one or more `drmd:material`

Best practices

Include **exactly one** `drmd:material` when the DRMD is for a single CRM/RM product.

Use **multiple** `drmd:material` entries when:

- The DRMD certifies multiple distinct materials.
- The product is a kit with distinct components.
- Variants/forms are treated as separate "materials" with separate sample size constraints or identifiers.

Keep identifiers stable across releases; if you revise the document, avoid changing identifiers unless the material itself changes.

```
<drmd:materials>
  <drmd:material>
  </drmd:material>
</drmd:materials>
```

4.3 Material – Each

Element: `drmd:material`
Path: `/drmd:digitalReferenceMaterialDocument/materials/drmd:material`
Type: `drmd:materialType`
Cardinality: 1..unbounded within `drmd:materials`

Name - **Path:** `.../drmd:material/drmd:name`
Type: `dcc:textType`
Cardinality: required (1)

Meaning: Human-readable **primary name** of the material (e.g., "BAM-M308a", "AlZnMgCu1.5", "CRM XYZ-123").

Purpose / use

- **Producer:** Public-facing product name.
- **Lab:** Selection of correct material in procedures and reports.
- **Software:** Display label and user selection (not a stable key—use identifiers for matching).

Best practices

- Provide at least one `dcc:content`.
- If multilingual usage is expected, include multiple `dcc:content` entries with `@lang`.
- Keep naming consistent with packaging/catalog.
- Provide one dcc:content if you only publish in one language.
- Provide multiple with @lang if the DRMD is intended for multilingual consumption.
- Keep the "material name" short; put extended marketing/narrative into drmd:description.

```
<drmd:name>
  <dcc:content lang="en">Aluminium Alloy Disk</dcc:content>
  <dcc:content lang="de">Aluminiumscheibe (Legierung)</dcc:content>
</drmd:name>
```

Description - **Path:** `.../drmd:material/drmd:description`
Type: `dcc:richContentType`
Cardinality: optional (0..1)

Meaning: Structured descriptive content: narrative text, attachments, formulas—typically used for form factor, composition notes, packaging form, hazards summary pointer, etc.

Best practices

- Use description to capture: physical form, dimensions, matrix, appearance, preparation hints.
- Keep long legal/h&s statements in `drmd:statements` (not here), but you may reference them.
- Use dcc:content for narrative.
- Use dcc:file for images/spec sheets only when needed (otherwise keep DRMD lean).

- Keep units/numerical constraints in itemQuantities or results sections rather than burying them only in narrative text.

Example (Text-only):

```
<drmd:description>
  <dcc:name>
    <dcc:content lang="en">Material description</dcc:content>
  </dcc:name>
  <dcc:content lang="en">
  The reference material is provided as discs (approx. 65 mm diameter, 30 mm height).
  Intended for establishing/checking calibration of spark OES and XRF methods for
      similar matrices.
  </dcc:content>
</drmd:description>
```

Example (with Attachment):

```
<drmd:description>
  <dcc:content lang="en">See the attached technical drawing for geometry and tolerances
      .</dcc:content>
  <dcc:file>
    <dcc:fileName>BAM-M308a-drawing.pdf</dcc:fileName>
    <dcc:mimeType>application/pdf</dcc:mimeType>
    <dcc:dataBase64>BASE64_PDF_BYTES_HERE==</dcc:dataBase64>
  </dcc:file>
</drmd:description>
```

4.4 Material classification (repeatable)

Material Class - Path: `.../drmd:material/drmd:materialClass`
Type: `drmd:materialClassType`
Cardinality: optional, repeatable (0..unbounded)

Content (inside `drmd:materialClassType`)

`drmd:reference` **(required):** `dcc:notEmptyStringType`

`drmd:classID` **(required):** `dcc:notEmptyStringType`

`drmd:link` **(optional):** `xs:anyURI`

Attributes (all optional): `@id` (xs:ID), `@refId` (xs:IDREFS), `@refType` (dcc:refTypesType).

Meaning: A classification entry that ties the material to a classification scheme and a class identifier within that scheme.

Purpose / use

- **Producer:** Align product with internal classification or external taxonomy.
- **Labs:** Group/filter materials by matrix/type (e.g., alloy class, soil class).
- **Manufacturers/software:** Enable automated method suggestions or UI filters by class.

Best practices

- Use **reference** for the scheme name (e.g., ISO, UNS, InternalCatalog, ECHA, etc.).
- Use **classID** for the scheme's code.
- Provide **link** when the scheme/class is resolvable online.

```
<drmd:materialClass id="mc-1">
  <drmd:reference>ISO 12103-1</drmd:reference>
  <drmd:classID>A2 Fine Test Dust</drmd:classID>
  <drmd:link>https://www.iso.org/standard/30109.html</drmd:link>
</drmd:materialClass>
```

4.5 Sample size and quantities (DCC list types)

4.5.1 Minimum Sample Size

Path: `.../drmd:material/drmd:minimumSampleSize`
Type: `dcc:itemQuantityListType`
Cardinality: required (1)

The minimum quantity of sample needed to use the material as intended (often depends on method family; if so, use multiple `dcc:itemQuantity` entries with names/descriptions inside the quantity structure).

Purpose / use

- **Labs:** Compliance with certified use conditions; planning sample prep.
- **Producer:** Communicate minimum amount for valid use.
- **Software/instruments:** Warnings if planned sample mass/volume is below minimum.

Best practices

- Use `si:real` for numeric sample size + unit (e.g., 0.2 g).
- If multiple minima exist (e.g., OES vs wet chemistry), provide multiple `dcc:itemQuantity` entries and differentiate using the quantity's name/description.

Example (minimum sample mass):

```
<drmd:minimumSampleSize>
  <dcc:itemQuantity>
    <si:real>
      <si:label>Minimum sample mass for wet chemical analysis</si:label>
      <si:value>0.2</si:value>
      <si:unit>\gram</si:unit>
    </si:real>
  </dcc:itemQuantity>
</drmd:minimumSampleSize>
```

Example (minimum sample size with uncertainty on the value):

```
<drmd:minimumSampleSize>
  <dcc:itemQuantity>
    <si:real>
      <si:label>Minimum sample mass</si:label>
      <si:value>0.200</si:value>
      <si:unit>\gram</si:unit>
      <si:measurementUncertaintyUnivariate>
        <si:expandedMU>
          <si:valueExpandedMU>0.005</si:valueExpandedMU>
          <si:coverageFactor>2</si:coverageFactor>
          <si:coverageProbability>0.95</si:coverageProbability>
          <si:distribution>normal</si:distribution>
        </si:expandedMU>
      </si:measurementUncertaintyUnivariate>
    </si:real>
  </dcc:itemQuantity>
</drmd:minimumSampleSize>
```

4.5.2 Item Quantities

Path: `.../drmd:material/drmd:itemQuantities`
Type: `dcc:itemQuantityListType`
Cardinality: optional (0..1)

Additional quantities describing the material item (not necessarily “minimum”), such as:

Nominal item mass, dimensions (diameter/height).

Number of units in package.

Storage temperature limits (if you prefer to keep numeric constraints close to the item).

Best practices

- Use this for **quantitative descriptors** of the material item/package.

- Avoid duplicating general statements that belong in `drmd:statements`.

Example (dimensions + mass as multiple `itemQuantity` entries):

```
<drmd:itemQuantities>
  <dcc:itemQuantity>
    <si:real>
      <si:label>Diameter</si:label>
      <si:value>65</si:value>
      <si:unit>mm</si:unit>
    </si:real>
  </dcc:itemQuantity>

  <dcc:itemQuantity>
    <si:real>
      <si:label>Height</si:label>
      <si:value>30</si:value>
      <si:unit>mm</si:unit>
    </si:real>
  </dcc:itemQuantity>

  <dcc:itemQuantity>
    <si:real>
      <si:label>Approximate mass</si:label>
      <si:value>250</si:value>
      <si:unit>\gram</si:unit>
    </si:real>
  </dcc:itemQuantity>
</drmd:itemQuantities>
```

Example (if you don't have a numeric quantity):

```
<drmd:itemQuantities>
  <dcc:itemQuantity>
    <dcc:noQuantity>
      <dcc:content lang="en">Item quantity not specified (varies by batch/packaging).</
          dcc:content>
    </dcc:noQuantity>
  </dcc:itemQuantity>
</drmd:itemQuantities>
```

4.6 Primitive Quantity Type

Type: `dcc:primitiveQuantityType`
Where used here: inside `dcc:itemQuantityListType` → `dcc:itemQuantity`

Structure (from DCC schema)
`dcc:primitiveQuantityType` contains:

`dcc:name` (optional, `dcc:textType`)

`dcc:description` (optional, `dcc:richContentType`)

Choice of one of:

- `dcc:noQuantity`, `dcc:charsXMLList`, `si:real`, `si:hybrid`, `si:complex`, `si:constant`, `si:realListXMLList`, `si:complexListXMLList`

Optional attributes: `@id`, `@refId`, `@refType`

A standardized container for a **single quantitative statement**, optionally with a name and description, using SI (D-SI) elements.

Purpose / use

Reference material producer (BAM, etc.): Publish minimum sample size, recommended test portion, mass/volume per unit, etc., in a consistent machine-readable structure.

Laboratories: Read constraints and compare them to method requirements; display them in LIMS/reporting systems.

Instrument manufacturers / measurement software: Parse numeric values + units reliably, validate completeness, and present values consistently in UIs, PDFs, and exports.

Schema/tooling developers: Validate that exactly one "quantity payload option" is chosen, and that required numeric/unit parts exist for the selected option.

Best practices

- Provide `dcc:name` whenever a list contains multiple quantities (humans and software need labels).
- Default to **Option A (si:real)** for numeric + unit (highest interoperability).
- Use **Option C (si:realListXMLList)** only when your ecosystem supports list parsing; otherwise model multiple `dcc:itemQuantity` entries each with `si:real`.
- Use **Option B (dcc:noQuantity)** only when numeric is truly not applicable.
- Use `dcc:charsXMLList` for controlled vocabulary / token lists—not for numbers.
- Treat `si:hybrid`, `si:complex`, `si:constant`, `si:complexListXMLList` as **advanced/rare** options.

4.6.1 Option A — si:real

Use when you have **one number** with **one unit**.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Minimum sample size</dcc:content></dcc:name>
  <si:real>
    <si:value>0.2</si:value>
    <si:unit>g</si:unit>
  </si:real>
</dcc:itemQuantity>
```

4.6.2 Option B — dcc:noQuantity

Use when the value is **intentionally not numeric**, but you want a structured placeholder and explanation.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Minimum sample size</dcc:content></dcc:name>
  <dcc:noQuantity>
    <dcc:content lang="en">Not specified; depends on method.</dcc:content>
  </dcc:noQuantity>
</dcc:itemQuantity>
```

4.6.3 Option C — si:realListXMLList

Use when one “quantity” is naturally a **list of numbers**, each paired with a unit entry.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Recommended test portions</dcc:content></dcc:name>
  <si:realListXMLList>
    <si:valueXMLList>0.1</si:valueXMLList>
    <si:valueXMLList>0.2</si:valueXMLList>
    <si:unitXMLList>g</si:unitXMLList>
    <si:unitXMLList>g</si:unitXMLList>
  </si:realListXMLList>
</dcc:itemQuantity>
```

4.6.4 Option D — dcc:charsXMLList

Use when the “quantity” is best represented as a **list of tokens/strings** (not numeric), e.g., categorical values, codes, labels.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Applicable methods</dcc:content></dcc:name>
  <dcc:charsXMLList>ICP-OES XRF OES</dcc:charsXMLList>
</dcc:itemQuantity>
```

4.6.5 Option E — si:hybrid

Use when you need a **hybrid** representation, typically when a plain double is insufficient.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Hybrid-valued quantity</dcc:content></dcc:name>
  <si:hybrid/>
</dcc:itemQuantity>
```

4.6.6 Option F — si:complex

Use when the quantity is genuinely **complex-valued** (real + imaginary).

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Complex-valued quantity</dcc:content></dcc:name>
  <si:complex/>
</dcc:itemQuantity>
```

4.6.7 Option G — si:constant

Use when you need to represent a **constant** quantity as defined by the SI schema.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Constant quantity</dcc:content></dcc:name>
  <si:constant/>
</dcc:itemQuantity>
```

4.6.8 Option H — si:complexListXMLList

Use when you need a **list of complex numbers** in a compact list form.

```
<dcc:itemQuantity>
  <dcc:name><dcc:content lang="en">Complex list quantity</dcc:content></dcc:name>
  <si:complexListXMLList/>
</dcc:itemQuantity>
```

4.7 Material Identifiers

Material Identifiers provide the **canonical, machine-actionable keys** to uniquely (or strongly) identify a reference material across producers, lots/batches, catalogs, and downstream systems. Unlike free-text names, identifiers are intended to be stable and suitable for indexing, deduplication, traceability, and cross-document linking.

This is especially important when:

- The producer offers multiple **forms** of the same RM (e.g., discs vs powder).
- Multiple **lots/batches** exist under the same catalog/product line.
- A DRMD contains multiple `drmd:material` entries.
- Software needs deterministic linking between "materials" and later "properties/results".

4.7.1 Material Identifiers List

Path: `.../drmd:materials/drmd:material/materialIdentifiers`
Type: `drmd:materialIdentifierListType`
Cardinality: optional (0..1)

Purpose / use

Producer: Publish official product codes and traceability anchors (catalog ID, lot/batch, internal ERP ID, DOI/Handle, etc.).

Laboratories / users: Verify the delivered material matches the documented material (especially by **lot/batch**).

Software developers: Use as primary keys for matching, deduplicating, syncing, and linking across systems (LIMS/ERP/PLM/registry), instead of fuzzy string matching on names/descriptions.

Best practices

- Provide this list whenever the material must be uniquely tracked (almost always recommended).
- Include at least one **product-level identifier** (e.g., catalog/material code) and one **lot/batch-level identifier** when the DRMD is specific to a batch/production run.
- Keep identifiers **exactly as printed** on packaging/certificate (case, punctuation, separators).
- Prefer persistent/resolvable identifier schemes (DOI, Handle, official producer page) whenever possible.

4.7.2 Material Identifier

Path: `.../drmd:materials/drmd:material/materialIdentifiers/drmd:materialIdentifier`
Type: `drmd:identifierType`
Cardinality: unbounded within the list (0..∞)

Each `drmd:materialIdentifier` is one identifier instance, expressed as a **scheme** + **value** pair, optionally with a **link** and optional referencing attributes.

Structure drmd:scheme (required)
Path: `.../drmd:materialIdentifier/drmd:scheme`
Type: `dcc:notEmptyStringType`
Cardinality: required (1)

Names the identifier system / namespace / issuing authority.

Examples of good scheme values:

- Producer-specific: BAM, NIST, ERM, LGC
- Internal enterprise systems: ERP, SAP, InternalMaterialID
- Commercial identifiers: CatalogNo, GTIN
- Batch identifiers: Lot, Batch
- Persistent identifiers: DOI, Handle, URL

Best practices: Use a controlled set of scheme names across your organization to avoid duplicates. If you publish externally, prefer well-known scheme names.

drmd:value (required)
Path: `.../drmd:materialIdentifier/drmd:value`
Type: `dcc:notEmptyStringType`
Cardinality: required (1)

The identifier string itself (the token that someone would type/search/scan).

Best practices: Store the value exactly as used operationally (labels, invoice, CoA). Do not embed human explanations here; this field is meant to be a clean key.

drmd:link (optional)
Path: `.../drmd:materialIdentifier/drmd:link`
Type: `xs:anyURI`
Cardinality: optional (0..1)

A resolvable URL for the identifier (product page, registry entry, landing page).

Best practices: Use `https://` URLs where possible. Prefer stable landing pages.

4.7.3 Attributes

These attributes exist on **each** `drmd:materialIdentifier` and are useful for internal cross-references, merges, and structured linking patterns.

@id (optional):

- **Type:** `xs:ID`

- **D**efines an internal XML identifier for this node.
- **When to use:** You want to reference this exact element via `@refId` or want stable anchors for processing pipelines.

@refId (optional):

- **Type:** `xs:IDREFS`
- **O**ne or more references to `@id` values elsewhere in the document.
- **Clarification:** `@refId` is a general XML mechanism to reference other nodes. It supports internal linking such as mapping identifiers or connecting imported/legacy IDs to canonical ones.

@refType (optional):

- **Type:** `drmd:refTypesType` (enum: `basic_certificateIdentification`, `basic_measuredValue`
- **C**lassifies the purpose/type of the reference.
- **Best practices:** Use only when you have a clear internal convention for how software interprets the type.

4.7.4 Multi-Material Scenario

When a DRMD contains **multiple** `drmd:material` entries (e.g., different lots, different physical forms), the identifiers declared under `/drmd:materials/drmd:material/materialIdentifiers` should be treated as the **canonical keys** for linking downstream information.

Best practice

- Ensure each `drmd:material` has at least one identifier that is unique within the DRMD (often a lot/batch ID or internal material ID).
- In later sections (e.g., Material Properties), software should use these identifiers as stable references for associating certified property sets with the correct material, preventing mix-ups, and enabling robust database reconciliation.

Implementation options

- **XML-native approach:** Assign `@id` on a chosen identifier element and reference it using `@refId` patterns where the schema permits.
- **Database approach:** Map `(scheme, value)` pairs into normalized keys and use them as foreign keys across imported DRMD content.

Example

```xml
<drmd:materialIdentifiers>
  <drmd:materialIdentifier id="matid_m308a"
    refId="mat_001" refType="basic_certificateIdentification">
    <drmd:scheme>BAM</drmd:scheme>
    <drmd:value>M308a</drmd:value>
    <drmd:link>https://www.bam.de/</drmd:link>
  </drmd:materialIdentifier>
</drmd:materialIdentifiers>
```

5 Material Properties

Path:
/drmd:digitalReferenceMaterialDocument/drmd:materialPropertiesList

The Material Properties chapter contains the **property values** associated with the material(s) defined in the Materials chapter. It provides the structured, machine-readable representation of:

- **Certified and non-certified (informative) property sets** (distinguished by `drmd:materialPropert`
- **Results grouped into logical sections** (e.g., “Mass fraction”, “Density”, “Sputter factor”).
- **Numeric values with units and uncertainties** using the SI (D-SI) structures (e.g., `si:real`, `si:realListXMLList`).
- **Property identifiers** for each quantity (`drmd:quantity/drmd:propertyIdentifiers`) so software can reliably interpret what a value refers to (e.g., element/compound/property codes).
- **Procedures (methods) and measurement metadata** that provide provenance and interpretation context (method references, traceability statements, conventions, validity notes, responsible authority, etc.).

Purpose

Reference Material Producer (RMP):

- Publishes the authoritative dataset of certified values and clearly separates informative values.
- Provides transparency: uncertainty conventions, measurement procedures, and metadata needed for defensible use and audits.
- Supports lifecycle management: updates, revisions, and consistent dissemination of values across channels.

Laboratories / end users:

- Retrieve the correct values for their intended use (certified vs informative).
- Understand uncertainty meaning (coverage factor/probability) and the measurement context needed for correct reporting and accreditation evidence.
- Use property identifiers to avoid ambiguity (especially for multi-analyte tables).

Instrument / machine manufacturers:

- Ingest property tables for calibration/verification workflows and instrument libraries.
- Use units and uncertainties for automated validation, warnings, and calculation logic.

Software developers / LIMS / ELN / data platforms:

  - Parse results into structured records and map quantities to analytes/properties using `propertyIdentifiers` (rather than relying on labels).
  - Use `@id/@refId` conventions (best practice) to link property blocks to the correct material definition when multiple materials exist.
  - Enable robust machine processing: filtering by `@isCertified`, rendering tables, exporting to JSON/CSV, and maintaining traceability.

Regulators / auditors / accreditation bodies:

  - Verify that certified values are clearly distinguished from informative values.
  - Confirm that measurement provenance, conventions, and traceability statements are recorded in a structured form suitable for review.

5.1 Material Properties List

Path:
`/drmd:digitalReferenceMaterialDocument/drmd:materialPropertiesList`

A required top-level list of **material property blocks** (`drmd:materialProperties`), each bundling:

- A semantic title (name)
- Optional context (description)
- Optional procedures/methods (procedures)
- Required results (results)
- Optional measurement metadata (measurementMetaData)

Purpose / use

- **RM Producer:** Publish certified and informative datasets in separate blocks; maintain governance.
- **Laboratories:** Identify the correct value set (certified vs informative) and retrieve uncertainty/model details.
- **Instrument & software vendors:** Reliably parse result structures and render tables; filter by certification status (`@isCertified`).

Best practices

- Use **multiple** `materialProperties` blocks to separate:
  - Certified vs informative values (`@isCertified`)

  – Different analytical methods
  – Different property families (composition vs density vs sputter factors)

- Assign `@id` to each block for deterministic references.

Example

```
<drmd:materialPropertiesList>
  <drmd:materialProperties>...</drmd:materialProperties>
</drmd:materialPropertiesList>
```

5.2 Material Properties Type

Path:
/drmd:digitalReferenceMaterialDocument/drmd:materialPropertiesList/drmd:materialProperties

Structure (sequence) + required/optional

`drmd:name` — required — `dcc:textType`

`drmd:description` — optional — `dcc:richContentType`

`drmd:procedures` — optional — `dcc:usedMethodListType`

`drmd:results` — required — `drmd:resultListType`

`drmd:measurementMetaData` — optional — `dcc:measurementMetaDataListType`

Attributes

`@id` (xs:ID) optional

`@refId` (xs:IDREFS) optional

`@refType` (dcc:refTypesType) optional

`@isCertified` (xs:boolean) optional

The "chapter subsection" that groups results into a coherent dataset (e.g., "Certified values").

Purpose / use (multi-perspective)

- **Producer:** Declare scope and certification status; attach method list and metadata that applies to all results here.
- **Labs:** Interpret whether a block is certified; understand coverage factor conventions or validity notes.
- **Software:** Treat one `materialProperties` as one UI section; store `@id` as dataset key.

Best practices

- Use `@isCertified="true"` for certified datasets; for informative use omit or set false.
- Use `drmd:description` for conventions that apply to the whole block (e.g., "U is expanded uncertainty, k=2...").
- Use `drmd:procedures` when the same method(s) apply broadly to the block.

Example

```xml
<drmd:materialProperties id="mp_certified" refType="producerProfileToken" isCertified=
    "true">
  <drmd:name>
    <dcc:content lang="en">Certified Values</dcc:content>
  </drmd:name>

<drmd:description>
  <dcc:name>
    <dcc:content lang="en">Uncertainty convention</dcc:content>
  </dcc:name>
  <dcc:content lang="en">U is the estimated expanded uncertainty with coverage factor k
      = 2 (~95%).</dcc:content>
</drmd:description>

<drmd:procedures>
</drmd:procedures>

<drmd:results>
</drmd:results>

<drmd:measurementMetaData>
</drmd:measurementMetaData>
</drmd:materialProperties>
```

5.3 Result List Type

Path:
`.../drmd:materialProperties/drmd:results`

Structure

Contains `drmd:result` (1..∞) of type `drmd:resultType`

A required list of results in this dataset.

Purpose / use

- **Producer:** Define multiple results such as "Mass fraction", "Density", "Sputter factor".
- **Labs:** Locate values grouped by semantic result headings.
- **Software:** Render each result as a section/table.

Best practices

- Use one `drmd:result` per logical table or property group.
- Add `@id` to results if you plan references (e.g., external exports).

Example

```xml
<drmd:results>
  <drmd:result>...</drmd:result>
</drmd:results>
```

5.4 Result Type

Path:
`.../drmd:results/drmd:result`

Elements (sequence)

`drmd:name` — required — `dcc:textType`

`drmd:description` — optional — `dcc:richContentType`

`drmd:data` — required — `drmd:dataType` (choice)

Attributes

`@id` optional

`@refId` optional

`@refType` optional (`drmd:refTypesType` enum: `basic_certificateIdentification` | `basic_measure`

A named result container whose payload is either:

- A single `drmd:quantity`, or
- A `drmd:list` of quantities.

Purpose / use

- **Producer:** Give result-level semantics and conventions if needed.
- **Labs:** Cite the result section name in reports.
- **Software:** Group quantities into one output table.

Best practices

- Put shared wording (e.g., uncertainty definition) at the `materialProperties description` level if it applies everywhere; only repeat here if it is result-specific.
- Use `@refType="basic_measuredValue"` if you want to label results as measured values in a controlled way (optional).

Example

```xml
<drmd:result id="res_massFraction" refType="basic_measuredValue">
  <drmd:name>
    <dcc:content lang="en">Mass fraction</dcc:content>
  </drmd:name>

  <drmd:description>
    <dcc:content lang="en">Values are given as percent by mass.</dcc:content>
  </drmd:description>

  <drmd:data>
  </drmd:data>
</drmd:result>
```

5.5 Data Type

Path:
`.../drmd:result/drmd:data`

Choice (exactly one)

`drmd:list` (`drmd:listType`) or

`drmd:quantity` (`drmd:quantityType`)

Purpose / use

- **Producer:** Use `quantity` for single-value properties; `list` for multi-row tables.

- **Software:** Determine display structure.

Example

Best practices

- Prefer `drmd:list` for compositional tables (many elements).
- Prefer `drmd:quantity` for single properties (density, sputter factor if alone).

```
<drmd:data>
  <drmd:list>...</drmd:list>
</drmd:data>
```

5.6 List Type

Path:
.../drmd:data/drmd:list

Structure
`drmd:quantity` (1..∞) of type `drmd:quantityType`

A DRMD list container for one or more quantities.

Purpose / use

- **Producer/Labs:** Represent tables (e.g., analyte list).
- **Software:** Render as rows; each row is one `drmd:quantity`.

Best practices

- Ensure each quantity has a clear `dcc:name` (e.g., “Al”, “Fe”, “Density”).
- Use `drmd:propertyIdentifiers` for robust machine mapping.

Example

```
<drmd:list>
  <drmd:quantity>...</drmd:quantity>
  <drmd:quantity>...</drmd:quantity>
</drmd:list>
```

5.7 Quantity Type

Path:
`.../drmd:data/drmd:quantity` or `.../drmd:list/drmd:quantity`

Definition
`drmd:quantityType` extends `dcc:primitiveQuantityType` (**Refer Chapter 4.6 for detail information about dcc:primitiveQuantityType**) and adds:

`drmd:propertyIdentifiers` (optional, 0..1) of type `drmd:propertyIdentifierListType`

What can appear here (schema-complete)
From `dcc:primitiveQuantityType` you can have:

`dcc:name` optional (`dcc:textType`)

`dcc:description` optional (`dcc:richContentType`)

Then exactly one of:

- `dcc:noQuantity`, `dcc:charsXMLList`, `si:real`, `si:hybrid`, `si:complex`, `si:constant`, `si:realListXMLList`, `si:complexListXMLList`

Plus attributes: `@id`, `@refId`, `@refType` (inherited from DCC type)

Then DRMD adds:

`drmd:propertyIdentifiers` optional

Purpose / use (multi-perspective)

- **Producer:** Represent the property value (and its uncertainty) and attach stable identifiers.
- **Labs:** Read numeric values, units, uncertainties; identify analyte/property.
- **Software/instruments:** Parse numeric values and use property identifiers for mapping and interoperability.

Best practices

- Use `si:real` for single values with unit.
- Use uncertainty fields inside `si:real` for certified values.
- Provide `drmd:propertyIdentifiers` for each analyte/property in tables.

Example

```xml
<drmd:quantity id="q_al" refType="propertyRow">
  <dcc:name>
    <dcc:content lang="en">Al</dcc:content>
  </dcc:name>

  <dcc:description>
    <dcc:content lang="en">Mass fraction of Aluminium.</dcc:content>
  </dcc:description>

  <si:real>
    <si:label>mass fraction</si:label>
    <si:quantityTypeQUDT>MassFraction</si:quantityTypeQUDT>
    <si:value>49.5</si:value>
    <si:unit>\percent</si:unit>
    <si:measurementUncertaintyUnivariate>
      <si:expandedMU>
        <si:valueExpandedMU>1.3</si:valueExpandedMU>
        <si:coverageFactor>2</si:coverageFactor>
        <si:coverageProbability>0.95</si:coverageProbability>
        <si:distribution>normal</si:distribution>
      </si:expandedMU>
    </si:measurementUncertaintyUnivariate>
  </si:real>

  <drmd:propertyIdentifiers>
    <drmd:propertyIdentifier id="pid_al_cas" refType="basic_measuredValue">
      <drmd:scheme>CAS</drmd:scheme>
      <drmd:value>7429-90-5</drmd:value>
      <drmd:link>https://commonchemistry.cas.org/detail?cas_rn=7429-90-5</drmd:link>
    </drmd:propertyIdentifier>
  </drmd:propertyIdentifiers>
</drmd:quantity>
```

5.8 Property Identifiers

5.8.1 List

Path:
.../drmd:quantity/drmd:propertyIdentifiers

Type / Optional:
drmd:propertyIdentifierListType (Optional 0..1)

Description:
A list of identifiers that identifies **what property this quantity represents** (e.g., element, compound, property kind).

Purpose / use

- **Producer:** Make "Fe mass fraction" machine-readable.
- **Labs:** Reduce ambiguity in multi-analyte tables.
- **Software/instruments:** Robust mapping to analyte registries / internal measurement models.

Best practices

- Always include at least one property identifier for table rows.
- Use stable schemes (CAS, element symbol, internal analyte ID).
- Provide `link` where resolvable.

5.8.2 Item

Structure:

`scheme` (required)

`value` (required)

`link` (optional)

Attributes: `@id`, `@refId`, `@refType` (optional)

Example

```
<drmd:propertyIdentifiers>
  <drmd:propertyIdentifier id="pid_fe" refType="basic_measuredValue">
    <drmd:scheme>CAS</drmd:scheme>
    <drmd:value>7439-89-6</drmd:value>
    <drmd:link>https://commonchemistry.cas.org/detail?cas_rn=7439-89-6</drmd:link>
  </drmd:propertyIdentifier>
</drmd:propertyIdentifiers>
```

5.9 Procedures & Measurement Metadata

5.9.1 Procedures

Path:
`.../drmd:materialProperties/drmd:procedures`

Type / Optional:
Optional (0..1)

Structure (schema complete for DCC used methods):
`dcc:usedMethodListType` contains `dcc:usedMethod` (1..∞), each `dcc:usedMethodType` contains:

`dcc:name` (required, `dcc:textType`)

`dcc:description` (optional, `dcc:richContentType`)

`dcc:norm` (optional, 0..∞)

`dcc:reference` (optional, 0..∞)

`dcc:link` (optional, 0..∞)

`dcc:usedMethodQuantities` (optional, 0..1) containing `dcc:usedMethodQuantity` (0..∞) of `dcc:primitiveQuantityType`

Attributes: `@id`, `@refId`, `@refType` (optional)

Purpose / use

- **Producer:** Demonstrate traceability and reproducibility of property assignment.
- **Labs:** Assess method equivalence and comparability.
- **Auditors:** Evidence of documented procedures.

Best practices

- Include method name and at least one norm/reference/link when available.
- Only use `usedMethodQuantities` for parameters that matter (e.g., instrument settings).

Example

```
<drmd:procedures>
  <dcc:usedMethod id="m1">
    <dcc:name><dcc:content lang="en">ICP-OES</dcc:content></dcc:name>
    <dcc:description>
      <dcc:content lang="en">Elemental analysis by inductively coupled plasma optical
          emission spectrometry.</dcc:content>
    </dcc:description>
    <dcc:norm>ISO 11885</dcc:norm>
    <dcc:reference>Wilken et al., J. Anal. At. Spectrom. (2003)</dcc:reference>
    <dcc:link>https://www.iso.org/standard/56153.html</dcc:link>

    <dcc:usedMethodQuantities>
      <dcc:usedMethodQuantity>
```

```
        <dcc:name><dcc:content lang="en">Integration time</dcc:content></dcc:name>
        <si:real>
          <si:value>5</si:value>
          <si:unit>s</si:unit>
        </si:real>
      </dcc:usedMethodQuantity>
    </dcc:usedMethodQuantities>
  </dcc:usedMethod>
</drmd:procedures>
```

5.9.2 Measurement Meta Data

Path:
`.../drmd:materialProperties/drmd:measurementMetaData`

Type / Optional:
Optional (0..1)

Structure (schema complete):
`dcc:measurementMetaDataListType` → `dcc:metaData` (1..∞) of type `dcc:statementMetaDataType`. `dcc:statementMetaDataType` includes many optional fields such as:

- name, description
- countryCodeISO3166_1 (0..∞)
- convention, traceable
- norm / reference / link (0..∞)
- declaration
- valid or validXMLList
- date, period
- respAuthority (`dcc:contactType`)
- conformity or conformityXMLList
- data (`dcc:dataType`)
- nonSIDefinition, nonSIUnit
- location (`dcc:locationType`)
- Attributes: `@id`, `@refId`, `@refType`

Purpose / use

- **Producer:** Attach structured declarations: traceability, conventions, validity.
- **Labs:** Interpret confidence statements, validity dates/periods.
- **Software:** Show metadata panel; enable filters like `traceable=true`.

Best practices (to keep it readable)

- Use only the metadata fields you truly need; avoid duplicating prose that is already in description text.
- Use `traceable` and `convention` consistently.
- If you use validity metadata, include `date` or `period` so it is actionable.

Example

```
<drmd:measurementMetaData>
  <dcc:metaData id="md1">
    <dcc:name><dcc:content lang="en">Uncertainty convention</dcc:content></dcc:name>
    <dcc:convention>Expanded uncertainty, k=2 (~95%)</dcc:convention>
    <dcc:traceable>true</dcc:traceable>
    <dcc:norm>ISO/IEC Guide 98-3 (GUM)</dcc:norm>
    <dcc:link>https://www.iso.org/standard/50461.html</dcc:link>

    <dcc:valid>true</dcc:valid>
    <dcc:date>2026-05-07</dcc:date>
    <dcc:period>P3Y</dcc:period>

    <dcc:respAuthority>
      <dcc:name><dcc:content lang="en">BAM Reference Materials</dcc:content></dcc:name>
      <dcc:eMail>sales.crm@bam.de</dcc:eMail>
      <dcc:location>
        <dcc:city>Berlin</dcc:city>
        <dcc:countryCode>DE</dcc:countryCode>
      </dcc:location>
    </dcc:respAuthority>

    <dcc:location>
      <dcc:city>Berlin</dcc:city>
      <dcc:countryCode>DE</dcc:countryCode>
    </dcc:location>
  </dcc:metaData>
</drmd:measurementMetaData>
```

5.10 Property Values from SI

5.10.1 si:real

Fields (schema)

`label` — optional

`quantityTypeQUDT` — optional

`value` — required double

`unit` — required string

Optional `measurementUncertaintyUnivariate/expandedMU` **with:**

`valueExpandedMU`, `coverageFactor`, `coverageProbability`, `distribution` (all optional)

Best practices

- For certified values, use uncertainty block consistently (k, probability).
- Use a consistent unit encoding convention across DRMDs.

Example

```
<si:real>
  <si:value>0.594</si:value>
  <si:unit>\percent</si:unit>
  <si:measurementUncertaintyUnivariate>
    <si:expandedMU>
      <si:valueExpandedMU>0.011</si:valueExpandedMU>
      <si:coverageFactor>2</si:coverageFactor>
      <si:coverageProbability>0.95</si:coverageProbability>
      <si:distribution>normal</si:distribution>
    </si:expandedMU>
  </si:measurementUncertaintyUnivariate>
</si:real>
```

5.10.2 si:realListXMLList

Fields

`valueXMLList` — unbounded doubles

`unitXMLList` — unbounded strings

Example

```xml
<si:realListXMLList>
  <si:valueXMLList>0.81</si:valueXMLList>
  <si:valueXMLList>0.82</si:valueXMLList>
  <si:unitXMLList>\one</si:unitXMLList>
  <si:unitXMLList>\one</si:unitXMLList>
</si:realListXMLList>
```

5.10.3 Advanced SI

The “advanced” ones (`si:hybrid`, `si:complex`, `si:constant`, `si:complexListXMLList`) are declared but have **empty complex types**, so any example will necessarily look “empty” and won’t demonstrate real usage.

6 Statements

Path:
/drmd:digitalReferenceMaterialDocument/drmd:statements

The **Statements** chapter contains the human-readable declarations and guidance that accompany a reference material and its certified/informative values. It captures, in a structured but narrative-friendly way:

- The **intended use** of the RM (what it is fit for).
- **Storage requirements**.
- **Handling and use instructions**.
- **Optional statements** such as commutability, metrological traceability, health & safety, subcontractors, legal notice, and links/references to certification reports.
- A repeatable **"catch-all" statement** element for additional important statements not covered by the named fields.

All statement elements are typed as `dcc:richContentType`, allowing multilingual text and optional attachments (PDFs/images) and formulas.

Purpose

Reference Material Producer (RMP):

- Publishes normative guidance on how to correctly use the RM and interpret values.
- Ensures legal and safety information is available alongside data.
- Supports compliance with ISO 17034 / ISO 17025 expectations for documented use conditions.

Laboratories / end users:

- Understand correct storage/handling so certified values remain valid.
- Decide suitability for their application (intended use, commutability).
- Access safety instructions and references needed for audits and internal SOPs.

Instrument / machine manufacturers:

- Use handling/storage constraints to prevent misuse in workflows (warnings, UI messaging).
- Support method templates (e.g., "intended for calibration of XRF for similar alloys").

Software developers / LIMS / ELN / document platforms:

- Render statements as a structured section in generated PDFs/HTML.
- Support multilingual presentation and attachment download (SDS, certification report).

  - Allow searching/filtering by statement category (e.g., "healthAndSafetyInformation").

Regulators / auditors / accreditation bodies:

  - Verify that declared intended use, storage, and handling instructions are present and consistent with certification claims.
  - Review traceability and legal notices as part of conformity assessment.

6.1 Statements List

Path:
`/drmd:digitalReferenceMaterialDocument/drmd:statements`

A **required** container holding all statement subsections in a fixed order.

Best practices

- Always provide the required statements (`intendedUse`, `storageInformation`, `instructionsForHandli` with clear, actionable text.
- Use optional subsections whenever applicable rather than overloading the generic `statement` catch-all.

Example

```
<drmd:statements>
  <drmd:intendedUse>...</drmd:intendedUse>
  <drmd:storageInformation>...</drmd:storageInformation>
  <drmd:instructionsForHandlingAndUse>...</drmd:instructionsForHandlingAndUse>
</drmd:statements>
```

6.2 Common content model

All statement fields share the same structure and capabilities.

Structure

`dcc:richContentType` supports:

Optional `dcc:name` (a short heading title inside the statement).

One or more content groups containing any mix of:

  - `dcc:content` (multilingual text via `@lang`).
  - `dcc:file` (embedded attachment via `dcc:byteDataType`).
  - `dcc:formula` (latex/mathml via `dcc:formulaType`).

Optional attributes: `@id`, `@refId`, `@refType`.

Best practices

- Use `dcc:content` for the main narrative (keep it scannable; prefer bullets in text).
- Use `dcc:name` only if you need an internal sub-heading (often not necessary because the DRMD element name already gives a heading).
- Use `dcc:file` for externally required documents (e.g., SDS PDF, certification report excerpt) when you must embed; otherwise use links in text.
- Provide `@lang` when you publish multilingual statements.

Example

```
<drmd:intendedUse>
  <dcc:content lang="en">Intended for calibration of spark-OES for aluminium alloys
     with similar matrix.</dcc:content>
  <dcc:content lang="de">Geeignet zur Kalibrierung von Funkenspektrometern (OES) fuer
     Aluminiumlegierungen aehnlicher Matrix.</dcc:content>
</drmd:intendedUse>
```

6.3 DRMD Statements

6.3.1 Intended Use

Path: `.../drmd:statements/drmd:intendedUse`
Type: `dcc:richContentType`
Required: Yes (1)

Explains what the RM is **intended** (and typically **not intended**) to be used for.

Purpose / use (multi-perspective)

- **Producer:** Define scope and prevent misuse.
- **Labs:** Determine fitness for purpose; align with SOPs.
- **Auditors:** See declared scope consistent with claims.

Best practices

- State "intended for ..." and add explicit exclusions if relevant ("not intended for ...").
- Mention matrix/type and typical measurement techniques if appropriate.
- Keep it consistent with the types of properties presented as certified.

Example

```xml
<drmd:intendedUse>
  <dcc:content lang="en">
  Intended for calibration and quality control of elemental analysis methods (XRF,
     spark-OES) for aluminium alloys of similar matrix. Not intended for use as a
     primary standard for gravimetric preparation.
  </dcc:content>
</drmd:intendedUse>
```

6.3.2 Commutability

Path: `.../drmd:statements/drmd:commutability`
Type: `dcc:richContentType`
Required: Optional (0..1)

Provides commutability information (especially relevant for clinical/biological materials, but can apply elsewhere).

Purpose / use

- **Producer/Labs:** Whether the RM behaves like real samples across methods.
- **Auditors:** Documentation of commutability claims (if any).

Best practices

- Include only when commutability has been evaluated or needs to be stated explicitly.
- If not assessed, state that clearly.

Example

```xml
<drmd:commutability>
  <dcc:content lang="en">
  Commutability has not been assessed for this material. Use is restricted to methods
     validated for this matrix.
  </dcc:content>
</drmd:commutability>
```

6.3.3 Storage Information

Path: `.../drmd:statements/drmd:storageInformation`
Type: `dcc:richContentType`
Required: Yes (1)

Storage conditions needed to maintain material integrity and validity of certified values.

Purpose / use

- **Producer:** Specify preservation requirements.
- **Labs:** Implement correct storage controls (temperature, humidity, light).
- **Software/instruments:** Display storage constraints to users.

Best practices

- Provide actionable ranges/conditions in text (temperature, dry storage, sealed container).
- If storage affects validity, align with `drmd:administrativeData/coreData/validity`.

Example

```
<drmd:storageInformation>
  <dcc:content lang="en">
  Store in the original sealed container at room temperature (15-25 degC) in a dry
      environment.
  Protect from corrosive atmospheres and avoid condensation.
  </dcc:content>
</drmd:storageInformation>
```

6.3.4 Instructions for Handling and Use

Path: `.../drmd:statements/drmd:instructionsForHandlingAndUse`
Type: `dcc:richContentType`
Required: Yes (1)

Instructions to ensure correct use: preparation, homogeneity considerations, minimum sample size references, surface cleaning, etc.

Purpose / use

- **Producer:** Ensure correct use conditions.
- **Labs:** Integrate into SOPs; avoid invalid measurement.
- **Auditors:** Verify that use constraints are stated.

Best practices

- Reference minimum sample size rules (from Materials chapter) in a human-readable way.
- Include "do / do not" instructions (e.g., do not polish beyond X, avoid contamination).

Example

```
<drmd:instructionsForHandlingAndUse>
  <dcc:content lang="en">
  Before measurement, clean the surface with ethanol and a lint-free tissue.
  Avoid abrasion that may alter surface composition. Use a minimum test portion
      consistent with the minimum sample size stated in the Materials section.
  </dcc:content>
</drmd:instructionsForHandlingAndUse>
```

6.3.5 Metrological Traceability

Path: `.../drmd:statements/drmd:metrologicalTraceability`
Type: `dcc:richContentType`
Required: Optional (0..1)

Description:
Narrative traceability statement(s): traceability to SI, reference standards, methods, calibration chains.

Purpose / use

- **Producer:** Document traceability claim.
- **Labs:** Support accreditation evidence.
- **Auditors:** Review traceability statements.

Best practices

- State traceability clearly ("traceable to SI through ...").
- If not fully traceable, state limitations.

Example

```
<drmd:metrologicalTraceability>
  <dcc:content lang="en">
  Certified values are metrologically traceable to the SI through calibrated mass
      standards and validated analytical methods as described in the certification
      report.
  </dcc:content>
</drmd:metrologicalTraceability>
```

6.3.6 Health and Safety Information

Path: .../drmd:statements/drmd:healthAndSafetyInformation
Type: dcc:richContentType
Required: Optional (0..1)

Safety guidance; may include references or attachments (e.g., SDS).

Purpose / use

- **Producer:** Communicate hazards and safe handling.
- **Labs:** Comply with workplace safety processes.
- **Auditors:** Confirm safety info is supplied.

Best practices

- Include key hazard statements and PPE guidance.
- If an SDS exists, link or attach it via dcc:file.

Example

```xml
<drmd:healthAndSafetyInformation>
  <dcc:content lang="en">
  Handle as metal alloy. Avoid inhalation of dust if machining. Use appropriate PPE.
  Refer to the attached Safety Data Sheet (SDS).
  </dcc:content>
  <dcc:file>
    <dcc:fileName>SDS_BAM-M308a.pdf</dcc:fileName>
    <dcc:mimeType>application/pdf</dcc:mimeType>
    <dcc:dataBase64>BASE64_BYTES==</dcc:dataBase64>
  </dcc:file>
</drmd:healthAndSafetyInformation>
```

6.3.7 Subcontractors

Path: .../drmd:statements/drmd:subcontractors
Type: dcc:richContentType
Required: Optional (0..1)

Narrative listing of subcontracted activities or contributing labs/organizations.

Purpose / use

- **Producer:** Transparency of production/certification process.
- **Auditors:** Evaluate competence chain.

Best practices

- List subcontractors and roles (e.g., homogeneity study, interlaboratory comparison).
- Keep personal data minimal; use organizational contacts if needed.

Example

```xml
<drmd:subcontractors>
  <dcc:content lang="en">
  Selected measurements were carried out by  subcontracted laboratories participating
      in the interlaboratory comparison.
  Details are provided in the certification report.
  </dcc:content>
</drmd:subcontractors>
```

6.3.8 Legal Notice

Path: `.../drmd:statements/drmd:legalNotice`
Type: `dcc:richContentType`
Required: Optional (0..1)

Description:
Legal disclaimers, liability limitations, permitted use, and rights.

Purpose / use

- **Producer:** Legal protection and conditions of use.
- **Labs/Users:** Understand restrictions.
- **Auditors:** Verify standard disclaimers present.

Best practices

- Keep text clear and jurisdiction-appropriate.
- Avoid duplicating long boilerplate; consider linking to a maintained policy page.

Example

```
<drmd:legalNotice>
  <dcc:content lang="en">
  The producer's liability is limited to replacement of the material. Use is subject to
      the producer's terms and conditions.
  </dcc:content>
</drmd:legalNotice>
```

6.3.9 referenceToCertificationReport

Path: .../drmd:statements/drmd:referenceToCertificationReport
Type: dcc:richContentType
Required: Optional (0..1)

Reference (and optionally an attachment) to the detailed certification report.

Purpose / use

- **Producer:** Connect DRMD summary to the underlying report.
- **Labs/Auditors:** Locate evidence and detailed methodology.

Best practices

- Provide a stable URL or attach the report as a file if policy requires.
- Include document identifiers (report number, DOI) in text.

Example

```
<drmd:referenceToCertificationReport>
  <dcc:content lang="en">
  Certification report: BAM-CRM-M308a-2026 (available at https://example.org/reports/
     BAM-CRM-M308a-2026).
  </dcc:content>
</drmd:referenceToCertificationReport>
```

6.3.10 Statement (Catch-all)

Path: .../drmd:statements/drmd:statement
Type: dcc:richContentType
Required: Optional (0..∞)

A repeatable "additional statements" slot. Use it when something important must be included, but there is no dedicated element among the named fields.

Purpose / use (multi-perspective)

- **Producer:** Ensures no critical statement is lost due to schema constraints.
- **Labs:** Receive additional constraints or clarifications.
- **Software:** Display in an "Additional statements" section.

Best practices

- Use this **only** for statements that are important and not covered elsewhere.
- Use `dcc:name` to provide a clear heading (recommended for catch-all entries).
- Avoid duplicating content from other statement fields.

Example

```xml
<drmd:statement>
  <dcc:name>
    <dcc:content lang="en">Homogeneity note</dcc:content>
  </dcc:name>
  <dcc:content lang="en">
  The certified values apply to the minimum sample size stated in the Materials section
     . Smaller test portions may increase uncertainty.
  </dcc:content>
</drmd:statement>
```

6.4 Attachments

6.4.1 Attach a file (Optional)

When to use

- SDS, certification report excerpts, handling posters, technical drawings.

Best practices

- Prefer linking in text unless embedding is required.
- Keep file size reasonable; avoid duplicating public documents if a stable URL exists.

Example

```
<drmd:healthAndSafetyInformation>
  <dcc:content lang="en">See attached SDS.</dcc:content>
  <dcc:file>
    <dcc:fileName>SDS.pdf</dcc:fileName>
    <dcc:mimeType>application/pdf</dcc:mimeType>
    <dcc:dataBase64>BASE64_BYTES==</dcc:dataBase64>
  </dcc:file>
</drmd:healthAndSafetyInformation>
```

6.4.2 Formula (Optional)

When to use

- Definitions, conversions, calculation notes that benefit from a formula representation.

Example

```
<drmd:statement>
  <dcc:name><dcc:content lang="en">Uncertainty definition</dcc:content></dcc:name>
  <dcc:formula>
    <dcc:latex>U = k \cdot u_c</dcc:latex>
  </dcc:formula>
</drmd:statement>
```

7 Comments and Documents

Path
comment: `/drmd:digitalReferenceMaterialDocument/drmd:comment`
document: `/drmd:digitalReferenceMaterialDocument/drmd:document`

What is covered here (summary)
This chapter contains two optional, document-level add-ons:

- `drmd:comment` - a simple free-text note field intended for short, non-structured remarks about the DRMD instance.
- `drmd:document` - an embedded binary file (`dcc:byteDataType`), typically used to attach a **PDF rendition** of the DRMD / certificate (or another authoritative document representation) directly inside the XML.

These elements are not part of Materials, Material Properties, or Statements; they apply to the DRMD as a whole.

Purpose

Reference Material Producer (RMP):

- Uses `document` to deliver a "human-readable" official PDF alongside the machine-readable XML (single package).
- Uses `comment` for brief editorial or operational notes that are not part of the normative sections.

Laboratories / end users:

- Can store one artifact (XML) while still having access to the official PDF without a separate download.
- Use `comment` for quick notes in archives (e.g., "Imported into LIMS on ...").

Instrument / machine manufacturers:

- Usually ignore `document` for calculations, but may display it as "view certificate PDF".
- May show `comment` in UI logs/debug screens.

Software developers / LIMS / ELN / document platforms:

- `document` provides a canonical embedded file to store, checksum, and present to users.
- `comment` is a simple string for operational notes; not ideal for structured or multilingual content.

Regulators / auditors:

- `document` can provide the signed/issued PDF representation as evidence, stored together with the structured XML.
- `comment` should not be used for legally relevant statements (use Statements chapter for that).

7.1 Comment

Path: `/drmd:digitalReferenceMaterialDocument/drmd:comment`

Schema

Type: `xs:string`

Cardinality: optional (0..1)

Description

A short, document-level free-text note. It is **not multilingual** (unlike `dcc:textType`) and does not support attachments.

Best practices

- Keep it short and non-normative (e.g., import notes, processing notes).
- Do **not** put required guidance here (intended use, storage, handling belong in Statements).
- Do **not** put machine-relevant identifiers here (use identifier lists in the proper chapters).

Example

```xml
<drmd:comment>
Imported into LIMS on 2026-05-07. PDF attachment included as drmd:document.
</drmd:comment>
```

7.2 Document

Path: `/drmd:digitalReferenceMaterialDocument/drmd:document`

Schema

Type: `dcc:byteDataType`

Cardinality: optional (0..1)

Description

An embedded file container, most commonly used to include a **PDF version** of the DRMD/certificate in **base64** encoding, together with filename and MIME type. This is a **document-level attachment** (applies to the whole DRMD), in contrast to attachments embedded inside specific statement fields via `dcc:richContentType`.

Structure Inside `drmd:document` (`dcc:byteDataType`):

- `dcc:fileName` **(required):** Original or recommended file name (e.g., `BAM-M308a-certificate.pdf`). Include extension matching MIME type.
- `dcc:mimeType` **(required):** File media type (e.g., `application/pdf`). Use standard MIME types.
- `dcc:dataBase64` **(required):** The file bytes encoded as base64. Embed the exact bytes of the authoritative PDF.
- `dcc:name` **(optional):** Multilingual label for the embedded document.
- `dcc:description` **(optional):** Short description of what the embedded file represents.

Optional attributes on `drmd:document`

`@id`: recommended if you reference the embedded document elsewhere via `@refId`.

`@refId`: references other IDs (if you use a linking convention).

`@refType` (`dcc:refTypesType`): only if you have a consistent internal meaning.

Best practices

- Use it to embed the **official human-readable representation** (usually PDF).
- Prefer **one** document here (the canonical one). Put supplementary documents (SDS, drawings) into the relevant chapter sections (often Statements) as attachments.
- Ensure the embedded PDF matches the content and version of the XML (avoid mismatched revisions).
- Consider size/performance: base64 increases size; if file is very large, some ecosystems prefer external linking instead.

Example

```
<drmd:document id="doc_pdf_certificate">
  <dcc:name>
    <dcc:content lang="en">Certificate PDF</dcc:content>
    <dcc:content lang="de">Zertifikat (PDF)</dcc:content>
  </dcc:name>

  <dcc:description>
    <dcc:content lang="en">
    Human-readable PDF rendition of this DRMD instance.
    </dcc:content>
  </dcc:description>
```

```xml
    <dcc:fileName>BAM-M308a-certificate.pdf</dcc:fileName>
    <dcc:mimeType>application/pdf</dcc:mimeType>
    <dcc:dataBase64>JVBERi0xLjQKJcTl8uXrp...BASE64_BYTES_HERE...==</dcc:dataBase64>
  </drmd:document>
```

8 Digital Signature

Path:
/drmd:digitalReferenceMaterialDocument/ds:Signature

The **Digital Signature** section allows the DRMD to carry one or more **XML Digital Signature** blocks (`ds:Signature`) at the document root. A digital signature is used to protect:

- **Authenticity:** Who issued/signed the document.
- **Integrity:** That the XML content has not been modified after signing.
- **Non-repudiation / auditability:** Depending on organizational policy and legal framework.

In the DRMD schema, `ds:Signature` is optional and repeatable, enabling multiple signatures (e.g., producer seal + responsible person signature).

8.1 Purpose

Reference Material Producer (RMP):

- Digitally seals the DRMD to make the XML an authoritative, tamper-evident certificate artifact.
- Supports controlled issuance and traceability for revisions.

Laboratories / end users:

- Verify that the DRMD came from the expected producer and was not altered.
- Store and present verifiable evidence in audits/accreditation processes.

Software developers / LIMS / ELN / repositories:

- Automate signature verification on import.
- Enforce policies: "only accept certified values if signature validates".

Regulators / auditors / accreditation bodies:

- Use signature verification as part of evidence review and trust establishment.

Instrument / machine manufacturers:

- Optionally verify signature to ensure calibration libraries are loaded from trusted sources.

8.2 Schema

Element: `ds:Signature`
Cardinality:

> `minOccurs="0"` → not required
>
> `maxOccurs="unbounded"` → multiple signatures allowed

Important note about your XMLDSig schema:
You are importing a **minimal stub** (`xmldsig-core-schema.xsd`) for offline validation. In that stub, `ds:SignatureType` allows **any content** (`xs:any`, `processContents="lax"`). Consequently, the XSD will **not enforce** the real XMLDSig structure (e.g., `SignedInfo`, `SignatureValue`, `KeyInfo`). This is intentional for offline validation, but it means:

- Schema validity ≠ cryptographic validity.
- A document can be "XSD-valid" while containing a non-functional signature block.

8.3 Recommended Signing Patterns

One signature by producer (recommended default)

> **Pattern:** Add exactly **one** `ds:Signature` created with the producer's organizational certificate (electronic seal). This is the simplest and most interoperable approach.
>
> **Why it is recommended:** Clear trust anchor ("issued by the producer"), easy verification policy for consumers, and minimal ambiguity.

Multiple signatures (when needed)
Use multiple `ds:Signature` blocks if you have a defined governance model, such as:

- Producer organizational seal AND signature by a responsible person (or multiple persons).
- Timestamping authority signature (if represented as a signature block in your ecosystem).
- **Best practice:** If multiple signatures exist, document your verification rule (e.g., "at least one producer seal must validate").

What to sign?

> **Sign the whole document (recommended):** The safest default is to sign the **entire DRMD root element** content so that materials, properties, statements, and attachments are all integrity-protected.
>
> **Include embedded documents carefully:** Signing the whole XML will also protect embedded PDFs (`drmd:document`). Note that this increases signature verification cost due to document size.

Operational best practice: If performance becomes an issue, consider keeping `drmd:document` external (linked) and sign a hash of it elsewhere (if supported) OR keep the embedded PDF and accept higher verification cost.

8.4 Stub vs Production XMLDSig

Offline validation (with stub)

- Use XSD validation to ensure `ds:Signature` is allowed and structure is correct.
- Do **not** treat success of XSD validation as evidence of authentic signing.

Production validation (real signature verification)

In production systems, signature validation must be cryptographic and policy-driven:

- Use a full XMLDSig library (and preferably canonical XML processing).
- Validate the signature cryptographically, the certificate chain/trust store, and the revocation status (CRL/OCSP).
- **Best practice policy statement:** "Accept DRMD as certified only if at least one trusted producer signature validates successfully."

8.5 Best practices

- Prefer **exactly one** producer signature unless multi-signature is required.
- Define and publish a **verification policy** (trusted certificates, timestamp requirements, etc.).
- Ensure signing happens after the document is finalized (avoid post-sign modifications).
- If you publish a PDF in `drmd:document`, ensure it corresponds to the same version as the XML and that both are included in the signed scope.

Example

```xml
<ds:Signature>
  <!-- Real XMLDSig content would be here, e.g. SignedInfo, SignatureValue, KeyInfo,
      etc. -->
</ds:Signature>
```

9 Interoperability Profiles

This chapter defines three interoperability profiles for DRMD documents. The DRMD schema is flexible (many optional fields and rich narrative content). For automated processing and consistent behavior across software platforms, **reference material producers and DRMD consumers MUST agree on a profile** (A, B, or C).

The profiles define which information is required to be present in a **machine-readable** form, and which information may remain human-readable text.

Profiles are cumulative:

- **Profile A** defines the minimum interoperable core.
- **Profile B** adds requirements typically needed for **certified** reference materials.
- **Profile C** adds requirements for **archival packaging** and integrity (embedded PDF and signature).

D-SI (Digital SI) note (applies to all profiles)
For maximum interoperability, it is recommended that numerical values, uncertainties, and units are expressed using the **Digital SI (D-SI)** conventions. In DRMD this primarily affects the content of `si:unit` (and related SI/DCC quantity structures).

The detailed unit-string syntax rules and validation approach are defined in **Chapter 11 (Units, Quantities & Uncertainty Policy)** and **Chapter 12 (Validation Rules)**. Profile-level requirements are summarized below.

9.1 Profile A – Minimal interoperable DRMD (machine-readable core)

Goal: Enable reliable automated ingestion (parsing and extraction) of essential RM information with minimal constraints.

Profile A – MUST (minimum fields a parser can rely on)

- **Root**
  - `/drmd:digitalReferenceMaterialDocument/@schemaVersion` (required by schema)
- **Administrative data**
  - `drmd:administrativeData/drmd:coreData/drmd:titleOfTheDocument`
  - `drmd:administrativeData/drmd:coreData/drmd:uniqueIdentifier`
  - `drmd:administrativeData/drmd:coreData/drmd:validity`
  - `drmd:administrativeData/drmd:referenceMaterialProducer/drmd:name`

  - `drmd:administrativeData/drmd:referenceMaterialProducer/drmd:contact`
  - `drmd:administrativeData/dcc:respPersons` (required by schema structure)

- **Materials**
  - `drmd:materials/drmd:material` (at least one)
  - `drmd:material/drmd:name`
  - `drmd:material/drmd:minimumSampleSize`
- **Material properties (values)**
  - `drmd:materialPropertiesList/drmd:materialProperties` (at least one)
  - `drmd:materialProperties/drmd:name`
  - `drmd:materialProperties/drmd:results/drmd:result` (at least one)
  - `drmd:result/drmd:name`
  - `drmd:result/drmd:data` containing either:
    * `drmd:data/drmd:quantity` OR
    * `drmd:data/drmd:list/drmd:quantity` (one or more)
- **Statements (operational requirements)**
  - `drmd:statements/drmd:intendedUse`
  - `drmd:statements/drmd:storageInformation`
  - `drmd:statements/drmd:instructionsForHandlingAndUse`

Profile A – SHOULD (strong recommendations for machine-readability)

Quantities

  - For numeric quantities, producers SHOULD use: `si:real` (single value) or `si:realListXMLList` (lists).
  - If `si:real` is used, it MUST contain (per your SI stub): `si:value` and `si:unit`.

D-SI unit quality

  - Producers SHOULD encode `si:unit` according to D-SI unit-string syntax rules (see Chapter 11).
  - For Profile A interoperability, documents SHOULD meet at least **D-SI "Silver"** class for unit expressiveness.

Numeric formatting

  - Decimal separator MUST be a dot (.).
  - NaN/INF (and other non-finite numeric representations) MUST NOT be used in numeric fields.

Profile A – MAY (optional)

- Identifiers lists (`documentIdentifiers`, `materialIdentifiers`, `organizationIdentifiers`).
- Additional statement sections (traceability, safety, etc.).
- `drmd:comment`, `drmd:document`, `ds:Signature`.

9.2 Profile B – Certified RM (recommended)

Goal: Support certified reference materials where users need certified values, uncertainty, traceability, and unambiguous property identification. Profile B includes all Profile A requirements plus the following:

Profile B – MUST / SHOULD

Certified flag

- Each `drmd:materialProperties` block SHOULD include `@isCertified`.
- At least one `drmd:materialProperties` block SHOULD have `@isCertified="true"` to represent certified values.

Machine-readable uncertainty for certified values

- For quantities representing certified values (within `@isCertified="true"`), when `si:real` is used, producers SHOULD include: `si:measurementUncertaintyUnivariate/si:expandedMU/s`
- Producers SHOULD additionally provide at least one of: `si:coverageFactor` and/or `si:coverageProbability`. (`si:distribution` MAY be provided).

D-SI unit quality (certified values)

- For certified values, producers SHOULD meet **D-SI "Gold" or "Platinum"** class for unit expressiveness.
- For certified values, prefixes SHOULD be avoided where feasible to reduce ambiguity and improve comparability (Platinum-style practice).

Unambiguous property identifiers

- For each `drmd:quantity` in result tables/lists, producers SHOULD include: `drmd:propertyIdenti` (one or more).
- `scheme` + `value` (required), `link` (optional).
- Recommended identifier schemes include well-known nomenclatures (e.g., CAS, InChI/InChIKey) or a stable producer/internal property code that is documented.

Traceability statement: Producers SHOULD populate: `drmd:statements/drmd:metrologicalTrac`

Certification report reference: Producers SHOULD populate: `drmd:statements/drmd:referenceT` (link or `dcc:file`).

9.3 Profile C – Archival / full package

Goal: Enable long-term preservation and offline use by packaging machine-readable XML + human-readable representation + integrity protection. Profile C includes all Profile B requirements plus the following:

Profile C – SHOULD / MUST (as a package policy)

- **Embedded human-readable document**
  - Producers SHOULD include `drmd:document` (`dcc:byteDataType`) embedding the PDF representation:
  - `dcc:fileName` (required), `dcc:mimeType` (SHOULD be `application/pdf`), `dcc:dataBase64` (required).
- **Embedded supporting attachments where relevant**
  - Producers MAY embed additional documents (SDS, reports, handling instructions) as `dcc:file` inside relevant `dcc:richContentType` elements.
- **Digital signature**
  - Producers SHOULD include at least one `ds:Signature`.
  - Consumers verifying Profile C in production SHOULD perform cryptographic signature verification (XSD validation alone is not sufficient because the XMLDSig schema used for offline validation may be stubbed).
- **Multilingual completeness (if multilingual content is provided)**
  - If multiple languages are used, producers SHOULD apply them consistently across key fields (material name, key statements, and main result headings) to avoid partial translations.
- **D-SI strictness for archival**
  - Profile C documents SHOULD be fully consistent with the D-SI unit-string policy and numeric formatting rules defined in Chapter 11 and Chapter 12.

9.4 Conformance language and testing

9.4.1 Normative keywords

MUST: required for profile conformance.

SHOULD: recommended for interoperability; deviations should be justified.

MAY: optional.

9.4.2 Testing approach (recommended)

- **XSD validation:** Validate against the DRMD XSD and imported schemas/stubs. Checks structural correctness.
- **Profile validation (business rules):** Apply additional checks according to Profile A/B/C requirements (e.g., certified blocks, uncertainty presence, propertyIdentifiers presence, D-SI unit-string policy checks).
- **Cryptographic verification (Profile C):** If signatures are present and Profile C is claimed, perform XML Digital Signature verification using a trusted certificate policy.

9.4.3 Recommended conformance report output

A conformance report SHOULD include:

- Declared profile (A/B/C)
- Pass/fail result
- List of violations with:
  - Severity (ERROR/WARNING)
  - XPath/path to the offending location
  - Short description of the issue and expected requirement

10 Cross-reference & Identifier Conventions

This chapter defines recommended conventions for identifiers and cross-references in DRMD. The DRMD schema provides identifier structures and optional XML linking attributes (`@id`, `@refId`, `@refType`), but it does not enforce a single linking model. Without agreed conventions, software systems may interpret the same DRMD differently.

The goal of this chapter is to ensure that:

- Materials can be identified reliably across systems.
- Properties (measurands/analytes) can be mapped unambiguously.
- Multi-material documents can be processed deterministically.
- Imports do not create duplicates in LIMS/instrument libraries.

10.1 Canonical keys (stable identifiers)

10.1.1 General rule

For any identifier element of type `drmd:identifierType`, the canonical machine key is:

$$\textbf{CanonicalKey} = (\textbf{scheme}, \textbf{value})$$

because these two fields are mandatory in the schema. Relevant locations include:

- `drmd:documentIdentifiers/drmd:documentIdentifier`
- `drmd:materials/drmd:material/drmd:materialIdentifiers/drmd:materialIdentifier`
- `drmd:administrativeData/drmd:referenceMaterialProducer/organizationIdentifiers/drmd:o`
- `drmd:quantity/drmd:propertyIdentifiers/drmd:propertyIdentifier`

10.1.2 Normalization (recommended)

To improve matching across systems, consumers SHOULD normalize:

- **scheme:** trim whitespace; compare case-insensitively (unless you publish case-sensitive schemes).
- **value:** trim whitespace; preserve original string; compare per scheme rules (some schemes may be case sensitive).

Producers SHOULD avoid:

- Using multiple spellings for the same scheme (e.g., "CAS", "cas", "CASRN").
- Placing human-readable descriptions inside `value` (keep `value` as a true identifier).

10.1.3 Optional link

`drmd:identifierType/link` (optional) SHOULD be used when there is a stable public resolver: DOI URL, ROR URL, CAS link, or a producer landing page. Consumers MAY use `link` for UI display or lookups, but MUST NOT rely on it for identity.

10.2 Linking multi-material documents (which material do properties apply to?)

10.2.1 Problem statement

In the provided DRMD schema, `drmd:materialProperties` does not contain an explicit "material reference" element. Therefore, a best-practice convention is needed for documents containing more than one `drmd:material`.

10.2.2 Recommended convention (simple and deterministic)

Convention A (recommended):
If a DRMD contains multiple materials, each `drmd:materialProperties` block SHOULD indicate the target material by including at least one `propertyIdentifier` that identifies the material scope, using an agreed scheme.

Convention B (preferred for implementations):

- Producers SHOULD include an `@id` on each `drmd:material`.
- Producers SHOULD include a matching reference to that material in the relevant `drmd:materialProper` by using `@refId` (IDREFS) on `drmd:materialProperties`.

Note: Even though your drmd:materialPropertiesType currently does not define @refId in the snippet provided, the general DRMD/DCC pattern uses @id/@refId widely. If you keep your schema as-is, you can implement this convention only where attributes exist. If you later extend materialPropertiesType to add @refId, this becomes the cleanest linking model.

If no explicit linking is present, consumers SHOULD assume:

- All `materialProperties` apply to the single material if there is exactly one `drmd:material`.
- If there are multiple materials and no linking convention is used, the document is ambiguous and SHOULD raise a validation warning.

10.2.3 Minimum requirement for multi-material documents

For documents with multiple materials, producers SHOULD ensure material scope is determined by:

- Explicit ID linking (preferred).
- Separate DRMD documents per material (fallback).
- A clearly documented naming convention in `materialProperties/name` (least preferred).

10.3 Property identifiers convention (measurand/analyte identity)

10.3.1 Purpose

Property identifiers allow software to map measured/certified values to: analytes (e.g., Pb, Cd), measurands (e.g., mass fraction), and standardized nomenclatures (CAS, InChI).

10.3.2 Where property identifiers live (schema)

Path: `drmd:quantity/drmd:propertyIdentifiers/drmd:propertyIdentifier`
Type: `drmd:identifierType` (fields: `scheme`, `value`, optional `link`)

10.3.3 Recommended rules

- For each property represented as a `drmd:quantity`, producers SHOULD provide **at least one** `propertyIdentifier`.
- For compositional tables, provide one `propertyIdentifier` per row.
- Consumers SHOULD treat **(scheme, value)** as the property key.

10.3.4 Recommended schemes (examples)

It is recommended to use recognized nomenclatures where possible:

- **CAS** (CAS Registry Number) for chemical substances.
- **InChI** or **InChIKey** for chemical identity.
- **Element symbol** (e.g., scheme: `IUPAC_element_symbol`, value: `Fe`).
- **Producer internal property ID** (if stable and documented).

10.4 Duplicate prevention & merge rules (imports)

10.4.1 Material deduplication (recommended)

Consumers SHOULD match materials using, in priority order:

1. `coreData/uniqueIdentifier` (document identity).
2. `materialIdentifiers` **canonical keys** (scheme, value).
3. Fallback: normalized `material/name` text (least reliable).

10.4.2 Property deduplication (recommended)

Consumers SHOULD match properties using:

- `propertyIdentifiers` **canonical keys** (scheme, value) (preferred).
- Fallback: quantity name text (if provided), but treat as non-unique.

10.4.3 Merge behavior (recommended)

If an incoming DRMD matches an existing record:

- Consumers SHOULD merge identifiers (union of identifier sets).
- Consumers SHOULD NOT overwrite certified values silently; if values differ, treat as a new version or a conflict requiring user confirmation.

10.4.4 Collision handling

If two different entities share the same **(scheme, value)** unintentionally:

- Consumers SHOULD flag this as a validation error.
- Producers SHOULD correct the identifiers to restore uniqueness.

11 Units, Quantities & Uncertainty Policy

This chapter defines implementation rules for encoding **quantities, units, and uncertainties** in a DRMD so that different producers and software systems interpret values consistently. The schema allows multiple quantity representations (DCC + SI types), but interoperability requires clear policy.

This best practice recommends using the **Digital System of Units (D-SI)** to express unit strings in a machine-readable and unambiguous form. D-SI rules below are derived from the D-SI specification and are intended to be applied to all DRMD documents, especially those claiming Profiles A/B/C.

11.1 Unit string policy for si:unit (D-SI unit strings)

11.1.1 Scope: where unit strings occur

In the provided schemas, a unit string occurs primarily in:

- `si:real/si:unit` (mandatory when `si:real` is used)
- `si:realQuantityType/si:unit`
- `si:realListXMLListType/si:unitXMLList` (unit for lists)

For machine-readability, all unit strings SHOULD follow the same policy.

11.1.2 Encoding and general syntax rules (D-SI)

- Unit strings MUST be **UTF-8** compatible.
- Identifiers for units and prefixes are ASCII, **lower case**, and **prefixed with a backslash** \(e.g., `\metre`, `\micro`).
- Units are combined by **concatenation** (sequence) to represent multiplication.
- Exponents are represented by the operator: `\tothe{EXPONENT}`
- Numerical values associated with units MUST use a decimal dot (.) as the separator.

Interpretation rule:
A D-SI unit term is built from three components:

1. **Prefix** (optional)
2. **BIPM unit identifier** (required per term)
3. **Exponent** (optional, applied using `\tothe{}`)

11.1.3 Allowed units (identifiers)

The following categories are used in D-SI. Producers SHOULD prefer SI base and derived units where feasible.

a) SI base units (examples) `\metre`, `\kilogram`, `\second`, `\ampere`, `\kelvin`, `\mole`, `\candela`

b) Derived SI units (examples) `\newton`, `\joule`, `\pascal`, `\watt`, `\volt`, `\ohm`, `\degreecelsius`

c) Non-SI units accepted for use with SI (examples) `\hour`, `\minute`, `\day`, `\tonne`, `\litre`, `\electronvolt`, `\arcminute`

d) Dimensionless `\one` (dimension “number”)

Note: The schema does not restrict the ***`si:unit`*** *string. Therefore, conformance to D-SI is a* ***best-practice policy requirement****, not an XSD requirement.*

11.1.4 Prefix policy (D-SI)

Prefixes represent decimal multiples/submultiples (base 10).

Rules:

- **Single prefix rule:** Each unit term MUST have no more than **one** prefix.
- **Double prefixes are forbidden:** e.g., `\milli\kilo`... MUST NOT be used.
- **Kilogram rule:** `\kilogram` MUST NOT be combined with an additional prefix.
- Multiples/submultiples must be formed using `\gram` with prefixes (e.g., `\milli\gram`, `\mega\gram`).
- `\gram` MUST NOT be combined with the prefix `\kilo`.
- Units such as `\one`, `\decibel`, `\day`, `\hour`, and `\degreecelsius` MUST NOT be combined with prefixes.
- If a unit has both a prefix and an exponent, the exponent applies to **both** (prefix and unit).

11.1.5 Exponent policy (D-SI)

Powers are expressed using: `\tothe{EXPONENT}`

- Exponent MUST be a decimal number string.
- Integer exponents allowed: 2, -3.

- Specific decimal exponents allowed (square roots): 0.5, +0.5, -0.5.
- No spaces are allowed inside the exponent braces.
- Exponents are forbidden for the unit `\one`.

11.1.6 Examples (D-SI unit terms)

Micrometre (μm): `\micro\metre`

Square metre (m^2): `\metre\tothe{2}`

Kilometre per hour (km/h): `\kilo\metre\hour\tothe{-1}`

Pascal (kg/(m·s^2)): `\kilogram\metre\tothe{-1}\second\tothe{-2}`

Ohm (Ω): `\metre\tothe{2}\kilogram\second\tothe{-3}\ampere\tothe{-2}`

Milliampere hour (mAh): `\milli\ampere\hour`

11.1.7 D-SI quality classes

To make profile expectations explicit, D-SI defines quality classes.

- **Platinum (Next Generation):** Only SI base units, `\one`, and specific time/angle units; **no prefixes**.
- **Gold (2030):** SI units with prefixes and derived SI units.
- **Silver (2024):** Adds non-SI units permitted for use with SI (e.g., `\litre`, `\tonne`).
- **Bronze (2020):** Includes outdated units from previous SI editions (e.g., `\bar`, `\angstrom`).

Recommended DRMD mapping (consistent with Chapter 9):

- **Profile A:** target **Silver** or better.
- **Profile B:** target **Gold/Platinum** for certified values.
- **Profile C:** be consistent with the chosen policy across the entire document (archival).

11.2 Numeric formatting & rounding

11.2.1 Numeric representation

- All decimal separators MUST be `.` (dot).
- Numeric fields MUST represent finite numbers (no NaN/INF).
- Scientific notation MAY be used where allowed by XML numeric types (e.g., `1.23E-6`), as long as it is finite.

11.2.2 Recommended rounding policy

The schemas do not define rounding rules. To reduce inconsistent displays and imports, producers SHOULD:

- Round reported values to a precision that reflects the uncertainty (commonly: uncertainty has 1–2 significant digits; value rounded to the same decimal place).
- Apply the same rounding policy consistently across the document (especially for composition tables).

Consumers SHOULD:

- Store the numerical values as provided (do not re-round silently on import).
- Display using local formatting rules while preserving the underlying value.

11.2.3 Units and scaling (avoid ambiguity)

To support comparison and merging:

- Producers SHOULD avoid unnecessary prefix changes between rows of the same table (e.g., mg/kg in one row, μg/g in another).
- For certified values, producers SHOULD prefer stable base/derived units, and avoid prefixes where feasible (Platinum-style).

11.3 Uncertainty rules

11.3.1 Where uncertainty is expressed (schema)

For `si:real` and `si:realQuantityType`, uncertainty is represented (optionally) as: `si:measurementUncertaintyUnivariate/si:expandedMU/` containing:

- `si:valueExpandedMU` (expanded uncertainty value)
- `si:coverageFactor` (k)
- `si:coverageProbability` (e.g., 0.95)
- `si:distribution` (string)

All fields in the uncertainty block are optional in the SI stub; therefore, the following are **policy rules**.

11.3.2 When to include uncertainty

- For **certified values** (Profile B/C certified blocks), producers SHOULD provide expanded uncertainty.
- For indicative/informative values, uncertainty MAY be provided if available.

11.3.3 Minimum recommended uncertainty set for certified values

When uncertainty is provided for a certified value:

- `valueExpandedMU` SHOULD be present.
- At least one of `coverageFactor` or `coverageProbability` SHOULD be present.
- `distribution` MAY be provided where it supports correct interpretation.

11.3.4 Interpretation guidance

- `valueExpandedMU` is interpreted in the **same unit** as the measured value.
- `coverageFactor` and `coverageProbability` should be consistent (e.g., k≈2 often corresponds to ∼95% under normal assumptions; provide both when possible).
- Use `distribution` to clarify assumptions (e.g., "normal", "rectangular").

11.4 Lists vs multiple quantities

DRMD supports both:

1. Lists of quantities (repeating `drmd:quantity` elements).
2. List value encodings (e.g., `si:realListXMLList`).

11.4.1 When to use si:real (single value)

Use `si:real` when each property value is a single scalar and you want:

- The cleanest numeric representation.
- Optional uncertainty per value.

Typical use: Single certified value per analyte/measurand.

11.4.2 When to use si:realListXMLList

Use `si:realListXMLList` when you have a **vector/list of numbers** that belongs together and is naturally processed as a list, such as:

- Series of calibration points.
- Replicated measurements.
- Time series values.
- Distribution samples.

Best practices: Maintain consistent list lengths and ensure the unit list (`unitXMLList`) is consistent and unambiguous.

11.4.3 When to use repeating drmd:quantity elements

Use repeating `drmd:quantity` when you have a **table** of different properties/analytes where each row is its own semantic entity and may carry:

- Its own `propertyIdentifiers`.
- Its own description/name.
- Its own uncertainty policy.

This is typically the best fit for **certified composition tables** and multiple measurands under one `materialProperties` block.

11.4.4 Consistency rules (recommended)

- Within one result table, producers SHOULD NOT mix list-encoded and row-encoded styles unless necessary.
- Prefer one consistent representation across the DRMD to simplify import logic.

12 Validation Rules & Severity Levels

This chapter defines validation requirements for DRMD documents beyond XML Schema (XSD) validation. XSD validation ensures that the document is structurally correct, but **structural correctness is not sufficient** to guarantee machine-readability, interoperability, or certification-grade completeness.

Validation is therefore split into three layers:

- **XSD validation (structure)** - verifies element/attribute presence, order, datatypes, and namespace correctness.
- **Profile validation (business rules)** - verifies that the document fulfills Profile A/B/C expectations (Chapter 9).
- **Cryptographic verification (signatures)** - verifies the authenticity and integrity of signed documents (Chapter 8).

The validation rules in this chapter are intended for:

- reference material producers (to validate before publishing),
- machine manufacturers/software vendors (to validate before importing),
- laboratories (to validate before using values for calibration/quality control),
- auditors (to evaluate completeness and traceability).

12.1 XSD validation requirements

12.1.1 What XSD validation MUST do

A DRMD validator MUST:

- validate the XML document against `drmd (6).xsd`,
- resolve and validate imports for:
  - `dcc (2).xsd`
  - `SI_Format.xsd`
  - `xmldsig-core-schema.xsd` (offline stub, if used)
- enforce `xs:sequence` ordering where required (order matters for many DRMD/DCC types),
- reject documents that violate required cardinalities or types (e.g., missing required elements).

12.1.2 What XSD validation does NOT guarantee

XSD validation does NOT guarantee:

- that textual fields are meaningful or non-empty (e.g., a required `dcc:richContentType` may technically exist but contain no useful content),
- that `si:unit` strings conform to D-SI rules (because `si:unit` is `xs:string`),
- that uncertainties are present for certified values (policy decision),
- that identifiers (scheme/value) use agreed vocabularies,
- that the digital signature is cryptographically valid.

12.1.3 XSD and XMLDSig stub limitation (critical)

When using the provided minimal `xmldsig-core-schema.xsd` stub, the schema allows:

- `ds:Signature` with arbitrary child elements (`xs:any`)

Therefore:

- A DRMD can be **XSD-valid** even if the embedded `ds:Signature` is missing required XMLDSig components or is cryptographically invalid.
- Any system that relies on signature validity MUST implement cryptographic verification independently of XSD.

12.2 Business rules (profile + interoperability rules)

Business rules are derived from: Profile expectations (Chapter 9), Identifier/linking conventions (Chapter 10), Units/uncertainty policy (Chapter 11), and practical RM use cases. Business rules SHOULD be evaluated after XSD validation succeeds.

12.2.1 Profile determination rule (recommended)

A validator SHOULD classify the document as the **highest profile it satisfies**:

- validate Profile A rules → then B → then C
- resulting conformance level is A, B, or C.

If the producer declares a target profile (e.g., via documentation, transport metadata, or packaging rules), then:

- failing the declared profile SHOULD be treated as an **ERROR**,
- passing a lower profile MAY be reported as **WARNING** ("downgraded to Profile A").

12.2.2 Core presence and non-empty content rules

The following elements are required by schema and SHOULD also be **meaningfully non-empty** for interoperability:

Required statements must be non-empty

- `drmd:statements/intendedUse`
- `drmd:statements/storageInformation`
- `drmd:statements/instructionsForHandlingAndUse`

Rule: each MUST contain at least one non-empty `dcc:content` string (after trimming whitespace). If present only as empty elements or whitespace-only content → **ERROR**.

Material and result naming must be non-empty

- Each `material/name` SHOULD contain at least one non-empty `dcc:content`.
- Each `materialProperties/name` and `result/name` SHOULD contain at least one non-empty `dcc:content`. Empty names create UI and mapping failures → **WARNING** or **ERROR** (depending on strictness).

12.2.3 Quantity encoding rules (machine-readability)

Preferred numeric encoding If a quantity is intended to be machine-usable numeric data, it SHOULD be expressed as: `si:real` or `si:realListXMLList`. Use of `dcc:noQuantity` or `dcc:charsXMLList` for certified/measured values SHOULD raise a **WARNING** (or **ERROR** in strict Profile B/C).

Unit required when numeric

- For `si:real`: `si:unit` is required by your SI stub; if missing → **ERROR**.
- For `si:realListXMLList`: `unitXMLList` is required by SI stub; if missing → **ERROR**.

12.2.4 D-SI unit string validation rules (critical interoperability)

Because `si:unit` is schema-typed as `xs:string`, a validator SHOULD implement D-SI checks.

Rule set:

- **UTF-8 / printable string check** (INFO/WARNING)
- **Backslash identifier format check** (ERROR for Profile A/B/C if required): unit term should be composed of \identifier tokens plus optional `\tothe{}` operators.
- **Prefix rules**
  - no double prefixes (ERROR)
  - no prefix with forbidden units (`\degreecelsius`, `\hour`, `\day`, `\one`, etc.) (ERROR)
  - no prefixing `\kilogram` (ERROR)
  - kg multiples must be built via `\gram` (WARNING/ERROR depending on policy)
- **Exponent rules**
  - exponent only via `\tothe{EXP}`
  - no spaces inside exponent braces (ERROR)
  - exponent forbidden for `\one` (ERROR)
  - allowed exponent forms: integers and ± 0.5 (WARNING if outside recommended set)
- **Medal class check**
  - Profile A: meets at least Silver (WARNING/ERROR depending on your profile enforcement)
  - Profile B: Gold/Platinum recommended for certified values (WARNING if below)
  - Profile C: consistent with defined policy (WARNING/ERROR)

Implementation note: if you do not implement a full D-SI grammar parser, start with regex + rule checks and classify failures conservatively as WARNING for Profile A, ERROR for Profile B/C certified values.

12.2.5 Numeric formatting and finite-number rules

Decimal separator Decimal separator MUST be dot (.). (**ERROR**). Most XML parsers enforce this already for `xs:double`, but validate explicitly if data originates from non-XML sources or conversions.

Finite numeric values Values MUST be finite (no NaN/INF). (**ERROR**). Again, many XML parsers will reject these for `xs:double`, but validate for safety in upstream conversions.

12.2.6 Certified values rules (Profile B/C)

These rules apply when `materialProperties/@isCertified="true"` is present, OR the producer/-consumer policy treats a given results block as certified.

Uncertainty presence For certified values represented by `si:real`, the uncertainty block SHOULD be present (Profile B/C). If missing → **WARNING** (Profile B) or **ERROR** (strict Profile B/C, depending on your policy).

Expanded uncertainty completeness (recommended)

- If `valueExpandedMU` is present, then `coverageFactor` and/or `coverageProbability` SHOULD be present. If neither is present → **WARNING** (or **ERROR** under strict certification policy).
- **Coverage probability range:** If provided, `coverageProbability` SHOULD be within (0, 1]. Out of range → **ERROR**.
- **Coverage factor positivity:** If provided, `coverageFactor` SHOULD be > 0. Non-positive → **ERROR**.
- **Uncertainty non-negativity:** `valueExpandedMU` SHOULD be ≥ 0. Negative → **ERROR**.

12.2.7 Property identifier rules (Chapter 10 consistency)

For list/table results

- If a `result/data/list` contains multiple `quantity` rows representing different analytes/properties: Each row SHOULD include at least one `propertyIdentifier` (**WARNING** for Profile A, **ERROR/WARNING** for Profile B depending on policy).
- `propertyIdentifier/scheme` and `value` are required by schema; ensure they are non-empty after trimming (**ERROR** if empty content is found via non-schema sources).
- **Scheme vocabulary:** If your ecosystem mandates specific schemes (CAS/InChI/etc.), validate scheme membership: unknown scheme → **WARNING** (or **ERROR** if strict).

12.2.8 Multi-material ambiguity rules

If `materials/material` appears more than once:

- The document SHOULD include an unambiguous mapping of each `materialProperties` block to a material (as defined by your Chapter 10 convention).
- If ambiguous → **WARNING** (Profile A) or **ERROR** (Profile B/C recommended).

12.2.9 Embedded document rules (drmd:document)

If `drmd:document` is present:

- `dcc:fileName`, `dcc:mimeType`, `dcc:dataBase64` are required by schema - missing → **ERROR**.
- `mimeType` SHOULD be consistent with `fileName` extension (**WARNING**).
- For `application/pdf`, decoded bytes SHOULD start with `%PDF-` (**WARNING/ERROR** depending on strictness).

12.2.10 Signature policy rules (Profile C)

If the document claims Profile C or contains `ds:Signature`:

- XSD validation alone is insufficient (**INFO** message).
- Cryptographic verification SHOULD be performed (**ERROR** if Profile C is required and verification fails, **WARNING** if verification is not performed).

12.3 Error severity taxonomy (ERROR / WARNING / INFO)

A DRMD validator SHOULD classify findings as follows:

12.3.1 ERROR (must not accept / must fail conformance)

Use **ERROR** when XSD validation fails, required elements are missing, operational statements are empty, numeric values are invalid, or strict Profile B/C policy requirements are violated.

> **Examples:** Missing `drmd:statements/storageInformation`, `si:real/si:unit` missing, `si:valueExp` is negative, `coverageProbability` = 1.5, D-SI string contains double prefix `\milli\kilo\gram`.

12.3.2 WARNING (acceptable but reduces interoperability/quality)

Use **WARNING** when the document is structurally valid but missing recommended fields, D-SI medal classes are low, or uncertainty info is incomplete.

> **Examples:** `materialIdentifiers` missing, `metrologicalTraceability` missing for Profile B, `propertyIdentifiers` missing in a composition list, certified block exists but lacks coverage info.

12.3.3 INFO (informational diagnostics)

Use **INFO** for additional helpful observations that do not indicate non-conformance.

> **Examples:** embedded PDF is large (base64 size warning), `ds:Signature` present but verification not performed (policy choice).

12.4 Suggested validation report format

A validation report SHOULD be machine-readable and should preserve exact locations for debugging.

12.4.1 Minimum report fields (recommended)

- **documentId:** extracted from `coreData/uniqueIdentifier` if present
- **schemaVersion:** root `@schemaVersion`
- **xsd_valid:** boolean
- **profile_target:** A/B/C (if declared)
- **profile_achieved:** A/B/C (highest passed)
- **signature_verification:** `not_present | not_checked | pass | fail`
- **issues:** list of issue objects

Each **issue** SHOULD include:

- **severity:** `ERROR | WARNING | INFO`
- **rule_id:** stable identifier (e.g., `BP12-STATEMENTS-NONEMPTY`)
- **message:** human-readable text
- **path:** XPath-like location (or JSON pointer equivalent)
- **context:** optional key/value (e.g., observed unit string)
- **profile:** which profile(s) the rule applies to (A/B/C)
- **category:** `xsd | profile | dsi | uncertainty | signature | identifiers | attachments`

12.4.2 Example JSON report (template)


```json
{
  "documentId": "BAM-CRM-M308a-2026",
  "schemaVersion": "0.2.0",
  "xsd_valid": true,
  "profile_target": "B",
  "profile_achieved": "A",
  "signature_verification": "not_present",
  "issues": [
  {
    "severity": "WARNING",
    "rule_id": "BP12-PROFILEB-TRACEABILITY-MISSING",
    "profile": "B",
    "category": "profile",
    "path": "/drmd:digitalReferenceMaterialDocument/drmd:statements/
       drmd:metrologicalTraceability",
    "message": "Profile B recommends providing a metrological traceability statement,
       but it is missing."
  },
  {
    "severity": "ERROR",
    "rule_id": "BP12-DSI-DOUBLEPREFIX",
    "profile": "A",
    "category": "dsi",
    "path": "/.../si:unit",
    "message": "Invalid D-SI unit string: double prefix is forbidden.",
    "context": { "unit": "\\milli\\kilo\\gram" }
  }
  ]
}
```


12.4.3 Tabular report alternative (human-readable)

If JSON is not feasible, a validator SHOULD emit a table with columns: **Severity** | **Rule ID** | **Profile** | **Path** | **Message** | **Observed Value**

13 Parsing & Data Extraction Guide

This chapter provides implementation guidance for software developers (machine manufacturers, LIMS/ELN vendors, data platforms) to reliably parse DRMD documents and extract the data needed for automated workflows (certificate verification, property extraction, calibration library import, archiving).

The DRMD schema builds on DCC and SI namespaces and uses structured text containers that support multiple languages and attachments. A robust parser must therefore handle:

- XML namespaces and prefix-independence,
- structured multilingual text (`dcc:textType`, `dcc:richContentType`),
- quantity payload choices (`si:real`, `si:realListXMLList`, etc.),
- optional linking via IDs and identifier lists,
- optional embedded PDF/attachments and signatures.

13.1 Namespace handling

13.1.1 Prefix-independence (critical)

XML prefixes are not significant. Do not assume the document uses the same prefix names as the schema examples. Your code **MUST match elements by namespace URI + local name**, not by prefix string.

Recommended namespace URIs (from your schemas):

DRMD: `https://example.org/drmd`

DCC: `https://ptb.de/dcc`

SI: `https://ptb.de/si`

XMLDSig: `http://www.w3.org/2000/09/xmldsig#`

13.1.2 Recommended parser configuration

- Use a namespace-aware XML parser.
- Disable external entity resolution (XXE protection).
- If validating, resolve schema imports from trusted locations (offline bundle or pinned versions).

13.1.3 Namespace map for XPath (example)

When using XPath libraries that require prefixes, define your own stable prefix map (regardless of the document's prefixes):

`drmd` → `https://example.org/drmd`

`dcc` → `https://ptb.de/dcc`

`si` → `https://ptb.de/si`

`ds` → `http://www.w3.org/2000/09/xmldsig#`

13.2 XPath/paths for common data (complete extraction map)

Notes:

- Paths below assume your local XPath prefix map (13.1.3).
- Where fields use `dcc:textType`, the actual human-readable string is in one or more `dcc:content` nodes.
- Many containers allow multiple languages; see normalization rules in 13.3.

13.2.1 Document root and schema version

Root element:
XPath: `/drmd:digitalReferenceMaterialDocument`

Schema version attribute:
XPath: `/drmd:digitalReferenceMaterialDocument/@schemaVersion`

13.2.2 AdministrativeData → CoreData

Title of the document
Enum value: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:coreD`

Unique identifier (document-level)
XPath: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:coreData/d`

Document identifiers (optional list)
Each identifier: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:coreData/drmd:documentIdentifiers/drmd:documentIdentifier`
Identifier parts:

- **scheme:** `./drmd:scheme/text()`
- **value:** `./drmd:value/text()`

– **link:** `./drmd:link/text()` (optional)

Validity
Validity container: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:coreData/drmd:validity`

– **Until revoked:** `./drmd:untilRevoked/text()`
– **Specific time:** `./drmd:specificTime/text()`
– **Time after dispatch:**
  * **dispatch date:** `./drmd:timeAfterDispatch/drmd:dispatchDate/text()` (optional)
  * **period:** `./drmd:timeAfterDispatch/drmd:period/text()` (required if timeAfterDispatch exists)

13.2.3 AdministrativeData → ReferenceMaterialProducer

Producer name (dcc:textType)
XPath (node): `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:referenceMaterialProducer/drmd:name`
Extract string(s): `./dcc:content/text()`
Recommended "primary display string": first matching language (see 13.3.1), else first content.

Producer contact
XPath: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:referenceM`

– **Contact name (dcc:textType):** `./dcc:name/dcc:content/text()`
– **Email, phone, fax:** `./dcc:eMail/text()` (optional), `./dcc:phone/text()` (optional), `./dcc:fax/text()` (optional)
– **Contact link:** `./dcc:link/text()` (optional)
– **Location (structured):** `./dcc:location/*` (various optional components; store as structured list)

Producer organization identifiers (optional)
Identifier list: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/drmd:refe` `drmd:organizationIdentifier`
scheme/value/link same as in previous sections.

13.2.4 AdministrativeData → Responsible persons (dcc:respPersons)

DRMD imports `dcc:respPersonListType`.

Persons list: `/drmd:digitalReferenceMaterialDocument/drmd:administrativeData/dcc:respPe`

Person details:

– `./dcc:person/dcc:name/dcc:content/text()`

- `./dcc:person/dcc:eMail/text()` etc.

Role: `./dcc:role/text()` (optional)

Main signer flags / crypto flags (optional):

- `./dcc:mainSigner/text()`
- `./dcc:cryptElectronicSignature/text()` etc.

13.2.5 Materials

All materials: `/drmd:digitalReferenceMaterialDocument/drmd:materials/drmd:material`

For each `drmd:material`:

- **Material name (dcc:textType)**: Node: `./drmd:name`, Strings: `./drmd:name/dcc:content/text()`
- **Material description (dcc:richContentType, optional)**: Node: `./drmd:description`, Text: `./drmd:description/dcc:content/text()`, Attached files: `./drmd:description/dcc:file` (see 13.2.10)
- **Material classification (repeatable)**: `./drmd:materialClass`, Fields: `./drmd:reference/text()`, `./drmd:classID/text()`, `./drmd:link/text()` (optional)
- **Minimum sample size (dcc:itemQuantityListType)**: `./drmd:minimumSampleSize`. This is a list of `dcc:itemQuantity` values; extract payload using 13.2.8.
- **Item quantities (optional)**: `./drmd:itemQuantities`
- **Material identifiers (optional list)**: `./drmd:materialIdentifiers/drmd:materialIdentifier` (scheme/value/link same as above).

13.2.6 Material Properties container and results

All materialProperties blocks: `/drmd:digitalReferenceMaterialDocument/drmd:materialPrope` `drmd:materialProperties`

For each `drmd:materialProperties`:

- **Certified flag**: Attribute: `./@isCertified` (optional; treat missing as "unknown")
- **Name (dcc:textType)**: `./drmd:name/dcc:content/text()`
- **Description (dcc:richContentType, optional)**: `./drmd:description/dcc:content/text()`
- **Procedures (optional)**: `./drmd:procedures` (type `dcc:usedMethodListType`)
- **Results list**: `./drmd:results/drmd:result`

For each `drmd:result`:

- **Result name**: `./drmd:name/dcc:content/text()`
- **Result description (optional rich content)**: `./drmd:description/dcc:content/text()`
- **Result data**: Node: `./drmd:data`, Choice: single: `./drmd:data/drmd:quantity` | list: `./drmd:data/d`

13.2.7 "Get all certified values" (common query)

Certified values are typically contained in materialProperties blocks with `@isCertified='true'`.

XPath (quantities under certified blocks):

```
/drmd:digitalReferenceMaterialDocument
/drmd:materialPropertiesList/drmd:materialProperties[@isCertified='true']
/drmd:results/drmd:result/drmd:data
/(drmd:quantity | drmd:list/drmd:quantity)
```

For each returned `drmd:quantity`, extract: value payload, unit(s), uncertainty, and propertyIdentifiers.

13.2.8 Quantity extraction (single unified algorithm)

`drmd:quantityType` extends `dcc:primitiveQuantityType`. Each `drmd:quantity` contains exactly one of the payload choices.

For each quantity, extract:

- **Optional label/name/description:** `./dcc:name` (dcc:textType), `./dcc:description` (dcc:richConte
- **Payload choice (exactly one):** `./si:real`, `./si:realListXMLList`, `./si:hybrid` / `./si:complex` / `./si:constant` / `./si:complexListXMLList` (present but stubbed), `./dcc:charsXMLList`, `./dcc:noQuantity`
- **If si:real:**
  - `./si:real/si:value/text()`
  - `./si:real/si:unit/text()`
  - **optional uncertainty:** `./si:real/si:measurementUncertaintyUnivariate/si:expandedMU/si:valueExpandedMU/text()`, `.../si:coverageFactor/text()`, `.../si:coverageProbability` `.../si:distribution/text()`
- **If si:realListXMLList:**
  - **values:** `./si:realListXMLList/si:valueXMLList/text()` (repeatable)
  - **units:** `./si:realListXMLList/si:unitXMLList/text()` (repeatable)
- **If dcc:charsXMLList:** `./dcc:charsXMLList/text()` (space-separated list)

- **If dcc:noQuantity:** treat as "present but non-numeric"; extract text from rich content: `./dcc:noQuantity/dcc:content/text()`
- **DRMD-specific property identifiers (important):** `./drmd:propertyIdentifiers/drmd:propert`
  Extract scheme/value/link.

13.2.9 Statements extraction

All statement elements are `dcc:richContentType`.
Statements container: `/drmd:digitalReferenceMaterialDocument/drmd:statements`
Common statement XPaths:

- **Intended use:** `.../drmd:statements/drmd:intendedUse`
- **Storage:** `.../drmd:statements/drmd:storageInformation`
- **Handling and use:** `.../drmd:statements/drmd:instructionsForHandlingAndUse`
- **Traceability:** `.../drmd:statements/drmd:metrologicalTraceability`
- **Catch-all statements:** `.../drmd:statements/drmd:statement`

For any `dcc:richContentType`, extract: text segments (`./dcc:content/text()` with optional `@lang`), attachments (`./dcc:file`), and formulas (`./dcc:formula`).

13.2.10 Attachments (dcc:byteDataType) extraction

Wherever `dcc:file` or `drmd:document` occurs:

- **fileName:** `./dcc:fileName/text()`
- **mimeType:** `./dcc:mimeType/text()`
- **base64 content:** `./dcc:dataBase64/text()`
- **optional name/description:** `./dcc:name/dcc:content/text()`, `./dcc:description/dcc:content`

Security best practice: treat decoded bytes as untrusted input; apply size limits and malware scanning.

13.2.11 Comment and embedded document (root-level)

- **Comment:** `/drmd:digitalReferenceMaterialDocument/drmd:comment/text()` (optional)
- **Embedded document (PDF or other):** `/drmd:digitalReferenceMaterialDocument/drmd:docum`
  Parse as `dcc:byteDataType`.

13.2.12 Digital signature(s)

Signature nodes: `/drmd:digitalReferenceMaterialDocument/ds:Signature` (0..n). Because the imported XMLDSig schema may be a stub, treat signature parsing as: store raw XML subtree, and verify cryptographically using an XMLDSig library.

13.3 Data normalization (recommended)

13.3.1 Language fallback for dcc:textType and dcc:richContentType

Many strings appear as `dcc:content` with optional `@lang` (ISO 639-1, two lowercase letters).

Recommended display selection algorithm:

1. If the application requests a language (e.g., `en`), select all `dcc:content[@lang='en']`.
2. Else if any `dcc:content` exists without `@lang`, use those as fallback.
3. Else use the first available `dcc:content` regardless of language.

Preserve all language variants in storage. For `dcc:richContentType`, keep multiple content blocks in order.

13.3.2 Unit normalization policy (D-SI)

- If your system uses D-SI internally: store `si:unit` as-is and validate per Chapter 11/12.
- If your system uses another unit model internally: parse D-SI string into a structured representation; store both original string and normalized internal object.
- Never "pretty print" and overwrite the original unit string.

13.3.3 Identifier normalization policy

For **(scheme, value)** keys: trim whitespace, normalize scheme spelling (Chapter 10), and keep raw values unchanged (some are case-sensitive). Store all identifiers as an array and designate one internal primary identifier.

13.3.4 Numeric normalization

Store as high-precision numeric type (double/decimal). Preserve original lexical form for exact round-trip output. Do not round on import; round only for display.

13.4 Recommended internal object model

A DRMD is best modeled as a small set of core entities.

13.4.1 Document model

DRMDDocument

- schemaVersion
- coreData (title, uniqueIdentifier, validity, documentIdentifiers)
- producer (name, contact, organizationIdentifiers)
- responsiblePersons[]
- materials[]
- propertySets[] (materialProperties blocks)
- statements (structured by section + catch-all list)
- comment?
- embeddedDocument? (byteDataType)
- signatures[] (raw XML + verification results)

13.4.2 Material model

Material

- name (multilingual)
- description? (rich content)
- materialClassifications[]
- minimumSampleSize[] (list of quantities)
- itemQuantities[] (optional list of quantities)
- identifiers[] (scheme/value/link)

13.4.3 Property set / results model

MaterialPropertySet (maps to `drmd:materialProperties`)

- name (multilingual)
- description?
- isCertified? (true/false/unknown)
- procedures[] (used methods, if provided)
- results[]
- measurementMetaData[] (if provided)

Result

- name (multilingual)
- description?
- data: either `quantity` (single) or `quantities[]` (list/table)

13.4.4 Quantity model

Quantity

- name? (multilingual)
- description? (rich content)
- payloadType (real, realList, charsList, noQuantity, other)
- value / values[]
- unit / units[]
- uncertainty? (expanded MU, k, probability, distribution)
- propertyIdentifiers[] (scheme/value/link)

13.4.5 Identifier model

Identifier

- scheme (string)
- value (string)
- link? (URI)

14 Security & Trust Model

The trust model ensures that the technical data (values, units, uncertainties) remains authentic and unaltered from the moment of issuance by the Reference Material Producer (RMP) to the moment of consumption by an analytical instrument or LIMS.

14.1 Trust Store & Certificate Governance

Digital signatures (defined in Chapter 8) are only as strong as the infrastructure supporting them.

- **Trust Anchors:** Consumers (laboratories and software) must maintain a "Trust Store" of authorized Root Certificates from recognized Reference Material Producers (e.g., BAM, NIST).
- **Identity Verification:** RMPs should use organizational certificates (Electronic Seals) issued by qualified Trust Service Providers (TSPs) to ensure the signature is legally linked to the institution.
- **Public Key Distribution:** Producers should publish their public keys or certificate fingerprints on official, secure websites (HTTPS) to allow software vendors to pre-configure trust.
- **Revocation Checks:** Software implementations should periodically check Certificate Revocation Lists (CRL) or use Online Certificate Status Protocol (OCSP) to ensure a producer's signing key hasn't been compromised.

14.2 Handling Embedded Documents

DRMD allows embedding binary files like PDFs and images. These are "untrusted" data streams and require strict security processing.

- **Size Limits:** To prevent Denial-of-Service (DoS) attacks via "XML bomb" or memory exhaustion, parsers should enforce a maximum size for `dcc:dataBase64` elements (e.g., 50 MB).
- **Malware Scanning:** All extracted attachments MUST be scanned for viruses and malicious macros before being opened by the end-user.
- **MIME-Type Whitelisting:** Systems should only process a restricted list of safe formats (e.g., `application/pdf`, `image/png`, `image/jpeg`). Executable files (e.g., .exe, .bat, .js) MUST be rejected.
- **Content Consistency:** The software should verify that the `dcc:fileName` extension matches the declared `dcc:mimeType` to prevent "extension spoofing".

14.3 Transport & Storage Integrity

Even with a digital signature, the operational environment must protect the DRMD from accidental or malicious corruption during its lifecycle.

- **Secure Transport:** DRMDs should always be exchanged over encrypted channels (e.g., HTTPS, SFTP, or TLS-encrypted API calls).
- **Storage Immutability:** Once a DRMD is imported into a LIMS or instrument library, it should be marked as "Read-Only." Any changes to the data must require the issuance of a new DRMD with a new `uniqueIdentifier`.
- **Hashing for Quick Verification:** Systems should generate and store a cryptographic hash (e.g., SHA-256) of the entire DRMD file upon import. This allows for rapid integrity checks without performing full XMLDSig verification every time the file is accessed.
- **Audit Trail Requirements:** All security-relevant events—such as signature verification success/failure, certificate expiration warnings, and attachment extractions—must be logged in a local system audit trail for regulatory review.

14.4 Summary of Security Responsibilities

Stakeholder	**Responsibility**
RM Producer	Signs document with a valid, non-expired organizational seal.
Software Developer	Implements automated signature validation and malware scanning.
Laboratory (User)	Maintains the trust store and reviews audit logs for signature failures.
Auditor	Verifies that the digital "Chain of Trust" is intact and policy-compliant.

15 Versioning, Change Management & Backward Compatibility

Managing changes effectively prevents "digital decay" where older certificates become unreadable by newer software, or newer certificates are rejected by legacy laboratory instruments.

15.1 Handling Unknown Schema Versions

Software MUST implement version checking by inspecting the `schemaVersion` attribute on the root element.

- **Major Version Mismatch:** If the software encounters a major version (e.g., $1.x.x$) higher than its internal logic supports, it SHOULD stop processing and alert the user. This prevents the misinterpretation of data that may have undergone structural changes.
- **Minor/Patch Version Mismatch:** If only the minor or patch version is higher (e.g., software supports 0.3.0 but the document is 0.3.5), the software SHOULD attempt to parse the document, as these changes are typically backward-compatible.
- **Strict vs. Lenient Modes:** It is recommended that parsers offer a "lenient mode" for archival viewing (reading what is possible) and a "strict mode" for automated calibration (ensuring 100% schema compliance).

15.2 Breaking vs. Non-Breaking Changes

The DRMD follows a semantic versioning logic to signal the impact of updates to developers.

Change Type	Version Increment	Description
Breaking	Major (e.g., $0.x$ to $1.x$)	Renaming mandatory elements, changing the data type of a value, or removing fields.
Non-Breaking	Minor (e.g., 0.3 to 0.4)	Adding optional elements or new identifier schemes.
Correction	Patch (e.g., 0.3.0 to 0.3.1)	Fixing typos in documentation or schema annotations without changing structure.

15.3 Migration Guidance

When a Reference Material Producer (RMP) updates a certificate or the underlying schema, the following operational rules apply:

- **Document Re-issuance:** Any change to the technical content (e.g., a revised certified value) requires a new `uniqueIdentifier` (UUID) to be issued.

- **Stable Identifiers:** Even if a document is updated to a new version, the `materialIdentifiers` (like a batch number or catalog ID) SHOULD remain unchanged to maintain the link to the physical material.
- **Upgrading Stored DRMDs:** When migrating a database of old DRMDs to a new schema version, software SHOULD preserve the original XML and the original digital signature as a "record of origin" before generating the transformed version.

15.4 Deprecation Policy

Fields or identifier schemes that are no longer recommended (e.g., an outdated unit string or a decommissioned database link) will enter a "Deprecation" phase.

- **Warning Period:** Deprecated fields will remain in the schema for at least one minor version cycle but will trigger a WARNING during Profile validation.
- **Removal:** Fields will only be removed in Major version updates.
- **Replacement:** Documentation MUST provide a clear mapping from the deprecated field to the new recommended field to assist developers in updating their parsers.

15.5 Summary for Stakeholders

- **Producers:** Always use the latest patch version for stability; never reuse a UUID for a modified document.
- **Developers:** Implement "Future-Proof" parsing that ignores unknown optional elements in minor version updates.
- **Laboratories:** Ensure your LIMS can store multiple versions of a certificate to maintain a complete audit trail.

16 Implementation Checklists (quick reference)

This chapter provides a final validation stage for all stakeholders. Use these checklists to ensure that a DRMD instance is not only structurally valid but also operationally ready for automated laboratory environments.

16.1 Producer Checklist

Ensure every digital certificate you publish meets these exact technical and regulatory requirements.

A. Document Identity & Administration

- [ ] **Schema Versioning:** The root element contains the `schemaVersion` attribute (e.g., "0.3.0") to indicate specification compatibility.
- [ ] **Document Title:** `titleOfTheDocument` is set to `referenceMaterialCertificate` for CRMs or `productInformationSheet` for informative RMs.
- [ ] **Unique Identifier:** A mandatory UUID is generated for each document to prevent global ID collisions.
- [ ] **Document Identifiers:** At least one external identifier (DOI, Batch ID) is included with a defined scheme and value.
- [ ] **Validity Period:** Exactly one method (fixed date, relative period, or "until revoked") is chosen to define the certificate's lifespan.
- [ ] **Producer Contact:** Mandatory name, email/phone, and structured location (including ISO `countryCode`) are provided.
- [ ] **Responsible Persons:** A list of accountable individuals is included, with at least one person flagged as the `mainSigner`.

B. Physical Material Definition

- [ ] **Material Identity:** Every material entry has a human-readable name and at least one canonical `materialIdentifier` (e.g., Catalog #).
- [ ] **Sample Constraints:** The mandatory `minimumSampleSize` is defined with a numeric value and D-SI unit.
- [ ] **Classification:** The material is tied to a formal scheme (e.g., ISO, UNS) using reference and `classID`.
- [ ] **Multi-Material Integrity:** If the document covers multiple materials, each is assigned a unique `@id` for downstream property linking.

C. Technical Property Data (The "Certified Value" Core)

- [ ] **Certification Flag:** Every `materialProperties` block explicitly declares `@isCertified="true"` for certified values or false for informative ones.
- [ ] **Certified Presence:** At least one block has `@isCertified="true"` to fulfill Profile B/C requirements.
- [ ] **Property Identifiers:** Every quantity row includes a `propertyIdentifier` (e.g., CAS #) to allow machine-mapping without relying on text labels.
- [ ] **D-SI Unit Quality:** Unit strings use backslash identifiers (e.g., `\gram`) and meet "Gold" or "Platinum" class for certified values.
- [ ] **Unit Prefix Policy:** For certified values, prefixes are avoided where feasible (Platinum-style) to ensure direct comparability.
- [ ] **Uncertainty Completeness:** Every certified value includes `valueExpandedMU` plus either a `coverageFactor` or `coverageProbability`.
- [ ] **Numeric Validation:** The `valueExpandedMU` is ≥ 0, the `coverageFactor` is > 0, and the `coverageProbability` is between $(0, 1]$.
- [ ] **Consistency:** A single result table does not mix list-encoded and row-encoded quantities.
- [ ] **Traceability:** A structured `measurementMetaData` block or narrative statement confirms metrological traceability to SI.

D. Statements & Archival

- [ ] **Mandatory Narrative:** `intendedUse`, `storageInformation`, and `instructionsForHandlingAndUse` contain meaningful, non-empty text.
- [ ] **Safety Info:** Hazards are documented in `healthAndSafetyInformation`, optionally with an attached SDS.
- [ ] **Embedded PDF:** Profile C documents include a Base64-encoded PDF version of the certificate in the document root element.
- [ ] **Digital Signature:** At least one `ds:Signature` is applied to protect the document's authenticity and integrity.

16.2 Software/LIMS Checklist

This checklist provides technical requirements for developers of Laboratory Information Management Systems (LIMS), Electronic Lab Notebooks (ELN), and analytical data platforms to ensure robust ingestion, validation, and storage of DRMD files.

A. Parser Configuration & Namespace Handling

[ ] **Namespace Awareness:** The parser MUST be namespace-aware and match elements by URI + local name, rather than assuming specific prefix strings.

[ ] **Prefix Independence:** Logic MUST NOT rely on document-provided prefixes (e.g., `drmd:`, `dcc:`, `si:`) as they are not significant.

[ ] **URI Mapping:** Maintain a stable internal map for the following URIs:

- **DRMD:** `https://example.org/drmd`
- **DCC:** `https://ptb.de/dcc`
- **SI:** `https://ptb.de/si`
- **XMLDSig:** `http://www.w3.org/2000/09/xmldsig#`

[ ] **XXE Protection:** External entity resolution MUST be disabled to protect against XML External Entity (XXE) attacks.

[ ] **Schema Resolution:** Resolve schema imports from trusted, pinned offline bundles rather than external URLs.

B. Validation & Conformance Layers

[ ] **Structural Check:** Perform XSD validation against `drmd.xsd` and all imported schemas (DCC, SI, XMLDSig stubs).

[ ] **Profile Determination:** Automatically classify the document as the highest profile (A, B, or C) it fully satisfies.

[ ] **Business Rule Enforcement:**

- Verify that mandatory narrative fields (intended use, storage, handling) are not whitespace-only.
- Confirm that all numeric values are finite (no NaN or INF).
- Check that `coverageProbability` is within $(0, 1]$ and `coverageFactor` is positive.

[ ] **D-SI Unit Validation:** Implement regex or grammar-based checks for D-SI unit strings, specifically flagging forbidden double prefixes (e.g., `\milli\kilo\gram`).

[ ] **Reporting:** Generate a machine-readable validation report (e.g., JSON) including pass/fail status, profile achieved, and specific issue XPaths.

C. Data Extraction & Normalization

[ ] **Unified Quantity Algorithm:** Implement a single extraction logic that handles all `dcc:primitiveQua` choices, prioritizing `si:real` and `si:realListXMLList`.

[ ] **Multilingual Selection:** Implement the designated fallback algorithm:

  - Select by requested application language (e.g., `en`).
  - Fall back to content without a `@lang` attribute.
  - Fall back to the first available content block.
- [ ] **Identifier Keys:** Normalize identifiers by trimming whitespace and using the (scheme, value) pair as the unique machine key.
- [ ] **Unit Preservation:** Store the original D-SI unit string as-is; never "pretty print" or overwrite the source string during import.
- [ ] **Numeric Precision:** Store values in high-precision types (double/decimal) and preserve the original lexical form to ensure round-trip integrity.

D. Security & Trust Management

- [ ] **Cryptographic Verification:** For Profile C, use a full XMLDSig library to verify signatures against a trusted certificate policy.
- [ ] **Attachment Safety:**
  - Enforce size limits on Base64-encoded binary data to prevent memory exhaustion.
  - Verify that `dcc:mimeType` matches the `dcc:fileName` extension.
  - Scan all extracted attachments for malware before presentation to users.
- [ ] **Version Handling:** Inspect `@schemaVersion` and implement strategy-based parsing (Strict for certification, Lenient for archival).

E. Data Model Mapping

- [ ] **Internal Objects:** Map the XML structure to a coherent internal object model consisting of:
  - **Document Model:** Version, core metadata, and signatures.
  - **Material Model:** Name, identifiers, and sample size constraints.
  - **Property Model:** Certified status, results, and measurement metadata.
  - **Quantity Model:** Values, units, uncertainty, and property identifiers.

Implementation Note: To ensure long-term data integrity, the system SHOULD NOT round numeric values on import; rounding should occur only for user display.

16.3 Instrument Vendor Checklist

This checklist is for manufacturers of analytical instrumentation, calibration software, and measurement systems that ingest DRMD certificates to automate method configuration and quality control.

A. Material Identification & Integration

- [ ] **Library Mapping:** Use the human-readable material name for user display while mapping the material to internal instrument libraries using stable `materialIdentifiers` (e.g., catalog codes, batch IDs).
- [ ] **Deterministic Linking:** Use material identifiers as canonical keys to ensure property data is associated with the correct physical sample, particularly in multi-material documents.
- [ ] **Traceability Verification:** Confirm that the digital document refers to a clearly identified material consistent with external references or physical labels.

B. Property Data Ingestion & Calibration

- [ ] **Certified Value Prioritization:** Filter property blocks by the `@isCertified` attribute to ensure only certified values are used for primary calibration anchors.
- [ ] **Robust Analyte Mapping:** Map measured results to internal measurands using `propertyIdentifier` (e.g., CAS numbers, element symbols) rather than relying on non-unique text labels.
- [ ] **Metrological Ingestion:** reliably parse numerical values, units, and uncertainties for automated validation, warnings, and calculation logic.
- [ ] **Uncertainty Interpretation:** Correctly interpret confidence levels (e.g., 95%) and coverage factors (k) provided in the `measurementUncertaintyUnivariate` block for reporting and accreditation evidence.
- [ ] **Property Families:** Group quantities into logical output tables based on property families like density, composition, or sputter factors.

C. Workflow & Operational Safety

- [ ] **Sample Size Enforcement:** Display the `minimumSampleSize` in the UI and trigger automated warnings if planned sample mass/volume is below the certified threshold.
- [ ] **Handling & Preparation Alerts:** Use storage and handling instructions to prevent misuse and provide surface cleaning or preparation reminders to the operator.
- [ ] **Method Templates:** Support automated method templates by ingesting standardized material metadata (e.g., "intended for calibration of spark-OES").
- [ ] **Operational Notes:** Display root-level comments in UI logs or debug screens for operational context.

D. Security, Trust & Documentation

- [ ] **Trust Anchor Verification:** Optionally verify digital signatures to ensure calibration libraries are loaded exclusively from trusted, authentic producer sources.

[ ] **Integrity Policy:** For Profile C documents, implement full cryptographic signature validation rather than relying on basic structural XSD checks.

[ ] **Official Representation:** Provide users access to the "official" human-readable PDF representation of the certificate via the root-level `drmd:document` element.

E. Data Normalization for Instruments

[ ] **Unit Normalization:** Parse D-SI unit strings (e.g., `\gram`) into internal units without overwriting or losing the original original unit string in storage.

[ ] **Language Selection:** Apply the recommended display selection algorithm to show results in the operator's preferred language.

[ ] **Numeric High-Precision:** Store values as high-precision numeric types (double/decimal) to ensure precision for automated calculations.

16.4 Auditor Checklist

This checklist is intended for accreditation bodies, regulatory auditors, and internal quality managers to evaluate whether a Digital Reference Material Document (DRMD) meets the requirements of ISO 17034 and the technical standards defined in this best practice.

A. Governance & Accountability

[ ] **Document Status:** Verify that the `titleOfTheDocument` correctly distinguishes between a Certified Reference Material (`referenceMaterialCertificate`) and an informative product (`productInformationSheet`).

[ ] **Unique Identification:** Confirm the presence of a mandatory `uniqueIdentifier` (UUID) for version tracking and prevention of ID collisions.

[ ] **Responsible Persons:** Verify that the `respPersons` list identifies who prepared, reviewed, and authorized the document.

[ ] **Role Authorization:** Ensure specific roles (e.g., "Quality Manager", "Technical Reviewer") are assigned to satisfy organizational governance and auditability.

[ ] **Electronic Signature Flags:** Check that cryptographic capability flags (signature/seal/-timestamp) for responsible persons align with the organization's issuance policy.

B. Traceability & Measurement Quality

[ ] **Certified vs. Informative:** Confirm that certified values are explicitly distinguished from informative values using the `@isCertified` attribute.

[ ] **Metrological Traceability:** Review the narrative `metrologicalTraceability` statement to ensure it documents the chain to the SI or appropriate reference standards.

- [ ] **Uncertainty Budgets:** Verify that all certified values include expanded uncertainty (`valueExpandedMU` along with a coverage factor (k) and/or coverage probability.
- [ ] **Measurement Metadata:** Check for structured declarations of uncertainty conventions, validity dates, and responsible authorities.
- [ ] **Assigned Procedures:** Ensure measurement procedures, norms (e.g., ISO), and references are cited to demonstrate the reproducibility of property assignment.
- [ ] **Subcontractor Transparency:** Verify that any subcontracted laboratory activities are documented within the `subcontractors` statement.

C. Operational Compliance

- [ ] **Lifespan & Expiration:** Verify the validity status and ensure that analytical instruments would trigger a "Hard Stop" if the material is past its expiration date.
- [ ] **Physical Linking:** Confirm that `materialIdentifiers` (batch/lot codes) provide a clear link between the digital record and the physical packaging.
- [ ] **Minimum Sample Size:** Ensure the `minimumSampleSize` is clearly stated and consistent with the certification claims.
- [ ] **Usage Instructions:** Review `intendedUse` and `instructionsForHandlingAndUse` to confirm they provide sufficient normative guidance to prevent misuse.
- [ ] **Certification Report:** Confirm a reference or stable link to the detailed certification report is included for evidence review.

D. Data Integrity & Trust (Profile C focus)

- [ ] **Authenticity Verification:** For documents claiming Profile C, verify the digital signature (`ds:Signature`) to ensure the XML content came from the stated producer and remains untampered.
- [ ] **Document Consistency:** Ensure the embedded PDF rendition in `drmd:document` matches the structured XML data in content and version.
- [ ] **Conformance Reporting:** Review the machine-readable validation report for any `ERROR` or `WARNING` flags that reduce interoperability or compliance.
- [ ] **Unit Standardization:** Confirm that units are expressed via the D-SI format to eliminate ambiguity in automated reporting.

Auditor Note: In the event of multi-material documents, verify that an unambiguous mapping exists (via `@id` or naming conventions) to associate properties with the specific material lot.

Appendix I – Schemas

This appendix provides the technical schema files and namespace definitions required for validating and parsing a Digital Reference Material Document (DRMD).

Appendix I.a DRMD Schema

`drmd.xsd` (Namespace: `https://example.org/drmd`).

Appendix I.b DCC Schema

`dcc.xsd` (Namespace: `https://ptb.de/dcc`).

Appendix I.c SI Schema

`SI_Format.xsd` (Namespace: `https://ptb.de/si`).

Appendix I.d XMLDSig Schema

`xmldsig-core-schema.xsd` (Namespace: `http://www.w3.org/2000/09/xmldsig#`).

[Full schema source code placeholders will be inserted here in the final version.]

Appendix II – Worked Examples (End-to-End)

This appendix contains complete, valid XML examples for common Reference Material scenarios, categorized by their interoperability profiles.

Appendix II.a Minimal DRMD (Profile A)

This example demonstrates the absolute minimum machine-readable core required for automated ingestion.

[Complete XML example for Profile A placeholder]

Appendix II.b Certified Composition DRMD (Profile B)

This example represents a standard Certified Reference Material (CRM).

[Complete XML example for Profile B placeholder]

Appendix II.c Multi-Material DRMD with Linking Conventions

This example demonstrates a complex document containing multiple materials using `@id` and `@refId` attributes.

[Complete XML example for Multi-material linking placeholder]

Appendix II.d Full Archival Package (Profile C)

The most comprehensive example, featuring a root-level embedded PDF and digital signatures.

[Complete XML example for Profile C placeholder]